
Catching a Moving Subspace: Low-Rank Bandits Beyond Stationarity

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Abstract

Many real bandit deployments (recommendation, clinical dosing, ad targeting) share two structural facts handled only in isolation by prior work: rewards live on a low-dimensional latent subspace, and that subspace drifts. Stationary low-rank bandits exploit the rank but break under any subspace change; non-stationary linear bandits adapt to drift but pay the ambient rate $\tilde{O}(d\sqrt{T})$. We study *piecewise-stationary low-rank* linear contextual bandits with scalar feedback: $\theta_t = B_k^* w_t$ for a rank- r factor $B_k^* \in \mathbb{R}^{d \times r}$ that is constant within each of K unknown segments and may shift at the boundaries. Our results are tight along three axes. **(i) Identification boundary.** With single-play scalar rewards, the moving subspace is recoverable through quadratic functionals of rewards if and only if three probe-side conditions hold (known noise variance, bounded state-noise coupling, full-dimensional probe support). Each is individually necessary in the unrestricted-second-moment problem, and the three are jointly sufficient: a characterization of the boundary of the solvable region, not just a sufficient interior point. **(ii) Algorithm and dynamic regret.** SPSC (Single-Play Subspace-Calibrated Optimism) interleaves isotropic probes with windowed projected ridge-UCB exploitation inside the learned r -dimensional subspace; a CUSUM-style adaptive variant discovers segment boundaries online. The costed dynamic regret is $\tilde{O}(r\sqrt{T}) + \tilde{O}(T^{2/3}) + O(W V_{\text{in}})$, replacing the ambient $d\sqrt{T}$ rate with the intrinsic rank. **(iii) Empirics.** On eleven benchmarks spanning synthetic, UCI/MovieLens, semi-synthetic clinical, and ZOZOTOWN production-log data, SPSC outperforms non-stationary and low-rank baselines whenever $d - r \gtrsim T^{1/6}$, matching the analytical crossover. To our knowledge, this is the first work to characterize the identification boundary and attain the intrinsic-rank dynamic-regret rate in this setting.

1 Introduction

High-dimensional sequential decisions — personalized recommendation, clinical dosing, ad allocation — share two structural facts that the linear-bandit literature has so far treated only in isolation: rewards live on a low-dimensional latent subspace, and that subspace drifts. User preferences, clinical responses, and ad-creative effects occupy a thin slice of the ambient $\mathbb{R}^d$ with intrinsic rank $r \ll d$; but the slice is *not static* — tastes drift, cohorts change, treatment regimes get revised, and the latent factor itself moves across unknown change points. Production deployments make both facts visible: Spotify’s homepage contextual bandit explicitly targets daily shifts in user intent [Feijer et al., 2025]; the IWPC warfarin cohort [Consortium, 2009] has been re-analyzed to expose three latent patient subgroups

whose dose-response coefficients flip sign across boundaries [Liu et al., 2025]; the public Open Bandit logs [Saito et al., 2021] ship scalar feedback collected across multiple ad campaigns at ZOZOTOWN.

Where the literature stops. Stationary low-rank bandits [Jun et al., 2019, Jedra et al., 2024, Lu et al., 2021, Jang et al., 2024] deliver the $\tilde{O}(r\sqrt{T})$ rate but assume a fixed factor; any non-trivial subspace change voids the analysis ($\tilde{O}$ suppresses polylog factors in $T, d, r, 1/\delta$). Non-stationary linear bandits [Russac et al., 2019, Cheung et al., 2019] absorb drift through sliding windows or discounting but operate in $\mathbb{R}^d$ and pay $\tilde{O}(d\sqrt{T})$ within each segment, ignoring intrinsic rank entirely. Naive combinations fail in instructive ways: sliding-window ridge in $\mathbb{R}^d$ never recovers the subspace; a one-shot subspace estimate breaks the moment a change point fires; per-segment re-runs of stationary low-rank algorithms require oracle access to the boundaries. No prior algorithm recovers a changing subspace under scalar feedback, and no prior analysis prices that recovery against the exploitation regret it buys.

This work. We study *piecewise-stationary low-rank* linear contextual bandits with scalar feedback: $\theta_t = B_k^* w_t$ for a rank- r factor $B_k^* \in \mathbb{R}^{d \times r}$ that is constant within each of K unknown segments $\mathcal{I}_k = [\tau_{k-1}, \tau_k)$ delimited by change points $1 = \tau_0 < \tau_1 < \dots < \tau_K = T+1$ and may shift at the boundaries, with the learner only observing scalar responses $y_t = x_t^\top \theta_t + \varepsilon_t$ (§2). The genuine difficulty is twofold. (a) A single scalar response y_t is a rank-one quadratic measurement of $\theta_t \theta_t^\top$, so the moving subspace is recoverable only through *second-order* probing. (b) Every probe diverts a round away from exploitation, so probe acquisition must itself be priced against the regret it enables. Our results are tight along three axes.

Contributions. (i) **Identification boundary (Theorem 2.2).** Three probe-side conditions (known noise variance, bounded state-noise coupling, full-dimensional probe support) are individually necessary in the unrestricted-second-moment problem and jointly sufficient for recovering $\text{range}(B_k^*)$ from quadratic functionals of scalar rewards. The necessity half — three matching impossibility results (Propositions B.12–B.14) — is what distinguishes this from a generic identifiability statement: it characterizes the boundary of the solvable region, not just a sufficient interior point. (ii) **Algorithm and costed dynamic regret (Theorem 4.1).** SPSC (Alg. 1) interleaves isotropic probes with windowed projected ridge-UCB exploitation inside the learned r -dimensional subspace via a quadratic-measurement identity; a **SPSC-Adaptive** variant (Alg. 2) uses a CUSUM-style detector to discover segment changes online. The costed dynamic regret is $\text{DynReg}_T^{(c)} \leq \tilde{O}(r\sqrt{T}) + \tilde{O}(T^{2/3}) + O(W V_{\text{in}})$, replacing the ambient $d\sqrt{T}$ rate with the intrinsic rank r , where W is the exploitation window and V_{in} the within-segment path variation. (iii) **Eleven benchmarks against eleven baselines.** A synthetic phase-transition grid, five UCI environments plus MovieLens, two semi-synthetic clinical datasets, the Russac small- d benchmark, and Open Bandit production logs, against LinUCB, D-LinUCB, SW-LinUCB, Restart-LinUCB, LowOFUL, VOFUL, LowRank-Reward, LinTS, SW-LinTS, and adapted stationary low-rank methods BOSS and Jedra. SPSC delivers regret reductions whenever $d - r \gtrsim T^{1/6}$, matching the analytical crossover. A direct necessity stress test (App. G.3, part C) restricts probe coverage to a proper subspace and reproduces the $\Omega(T)$ blow-up predicted by Proposition B.14, confirming the third identifiability condition is not a proof artifact.

To our knowledge, this is the first work to characterize the identification boundary and to attain the intrinsic-rank dynamic-regret rate in the piecewise-stationary low-rank setting under scalar feedback.

Related work. Stationary low-rank bandits [Jun et al., 2019, Kang et al., 2022, Jedra et al., 2024, Jang et al., 2024, Stojanovic et al., 2023, Duong et al., 2024, Lu et al., 2021] achieve intrinsic-rank rates but assume a fixed B^* ; non-stationary linear bandits [Russac et al., 2019, Cheung et al., 2019, Abbasi-Yadkori et al., 2023, Hou et al., 2024, Kim et al., 2022] bound dynamic regret in the ambient dimension and ignore low-rank structure; cost-aware observation [Seldin et al., 2014, Tucker et al., 2023, Elumar et al., 2024] prices information acquisition for fixed parameters, not for identifying a changing subspace. SPSC

sits at the intersection of all three: it (i) attains regret scaling with intrinsic rank r , (ii) adapts to changing subspaces without oracle boundaries, and (iii) prices probe acquisition explicitly. Appendix A gives the full positioning (Tables 5–7).

2 Setting and probe-based identification

Piecewise low-rank model. At every round t the reward parameter admits a factorization $\theta_t = B_k^* w_t$, where $B_k^* \in \mathbb{R}^{d \times r}$ has orthonormal columns and is piecewise constant over $K \geq 1$ segments $\mathcal{I}_k$ with unknown change points $1 = \tau_0 < \tau_1 < \dots < \tau_K = T+1$, and the latent state $w_t \in \mathbb{R}^r$ follows a stable linear dynamical system inside each segment

$$w_t = A_k w_{t-1} + \eta_{t-1}, \quad t \in \mathcal{I}_k,$$

with unknown A_k satisfying $\rho(A_k) \leq 1 - \alpha_0$ for some $\alpha_0 > 0$ (used only to ensure $\|w_t\| \leq S_w$ via the stationary covariance bound) and zero-mean innovation η_{t-1} with covariance $\Sigma_{\eta,k} \succ 0$. We assume uniform action-boundedness $\sup_{x \in \mathcal{A}_t} \|x\| \leq R_{\mathcal{A}}$, a known probe distribution Q with $\mathbb{E}[uu^\top] \succeq \rho_Q I_d$ and sub-Gaussian marginals (bounded $\|u\| \leq L$ is a special case; see Remark 2.1), conditionally σ_ε -sub-Gaussian noise with unknown variance, bounded state-noise coupling $\|\mathbb{E}[\varepsilon_t \theta_t]\|_2 \leq \epsilon_\times$, and full-dimensional probe coverage. Theorem 2.2 below shows these three probe-side conditions are individually necessary and jointly sufficient for identifying the changing subspace from single-play scalar rewards. Throughout, $\delta \in (0, 1)$ denotes the global confidence parameter (concentration bounds hold with probability $\geq 1 - \delta$) and $\lambda > 0$ denotes the ridge regularizer used by the windowed estimator of §3.

Remark 2.1 (Probe distribution: scaled sphere). *The theory uses scaled-sphere probes $u_t = \sqrt{d} v_t$, $v_t \sim \text{Unif}(\mathbb{S}^{d-1})$, which satisfy $\|u_t\|_2 = \sqrt{d}$ exactly (so no truncation event is needed) and $\mathbb{E}[uu^\top] = I_d$. In our implementation we draw $\tilde{u} \sim \mathcal{N}(0, I_d)$ and rescale $u \leftarrow \sqrt{d} \tilde{u} / \|\tilde{u}\|$, which is exactly the scaled-sphere distribution. The same argument applies to isotropic Gaussian probes $u_t \sim \mathcal{N}(0, I_d)$ via a χ_d^2 truncation at $L = \sqrt{2d \log(2T/\delta)}$ (Appendix C).*

Quadratic measurement identity. Let $\hat{\sigma}^2$ be the algorithm’s estimate of the noise variance and define the centered probe statistic $s_t := y_t^2 - \hat{\sigma}^2$ on every probe round $t \in \mathcal{T}_{\text{probe}}$. Under the probe-noise assumptions of §2, for $\delta_\sigma := \hat{\sigma}^2 - \sigma_\varepsilon^2$,

$$\mathbb{E}[s_t \mid \mathcal{H}_{t-1}, u_t] = u_t^\top \widetilde{M}_t u_t + 2u_t^\top \mathbb{E}[\varepsilon_t \theta_t \mid \mathcal{H}_{t-1}, u_t] - \delta_\sigma, \quad (1)$$

where $\widetilde{M}_t := \mathbb{E}[\theta_t \theta_t^\top \mid \mathcal{H}_{t-1}]$ is the predictable second moment, whose range is contained in $\text{range}(B_k^*)$ for $t \in \mathcal{I}_k$; under the non-degenerate innovation condition $\Sigma_{\eta,k} \succ 0$ (§2), the probe-time average has range exactly $\text{range}(B_k^*)$ (Proposition B.9). Under exact probe conditions ($\delta_\sigma = \epsilon_\times = 0$) this collapses to $\mathbb{E}[s_t \mid \cdot] = u_t^\top \widetilde{M}_t u_t$: a scalar reward carries a quadratic measurement of $\widetilde{M}_t$.

Lifted subspace estimator. The probe moment operator $\mathcal{K} : M \mapsto \mathbb{E}[(u^\top M u) u u^\top]$ admits a closed-form inverse on symmetric matrices for scaled-sphere probes $u = \sqrt{d} v$, $v \sim \text{Unif}(\mathbb{S}^{d-1})$:

$$\mathcal{K}(M) = \frac{d}{d+2} (\text{tr}(M) I_d + 2M), \quad \mathcal{K}^{-1}(N) = \frac{d+2}{2d} N - \frac{\text{tr}(N)}{2d} I_d, \quad \|\mathcal{K}^{-1}\|_{\text{op} \rightarrow \text{op}} \leq 1 \quad (2)$$

on the symmetric subspace (Lemma B.4). Define the *lifted probe sample*

$$G_t := \mathcal{K}^{-1}(s_t u_t u_t^\top), \quad \mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t + \widetilde{B}_t, \quad (3)$$

with bias $\widetilde{B}_t = -(\delta_\sigma/d) I_d$ exactly on scaled-sphere probes, hence $\|\widetilde{B}_t\|_{\text{op}} = |\delta_\sigma|/d$ (Lemma B.6). Averaging lifted samples across the probe rounds of segment k yields $\widehat{M}_k := m_k^{-1} \sum_{t \in \mathcal{T}_k} G_t$ (with $\mathcal{T}_k := \mathcal{T}_{\text{probe}} \cap \mathcal{I}_k$ the probe rounds in segment k , $m_k := |\mathcal{T}_k|$), whose top- r eigenvectors form $\widehat{U}_k \in \mathbb{R}^{d \times r}$ with $\|\widehat{P}_k - P_k^*\|_{\text{op}} \leq C_{\text{sub}} \sqrt{\log(d/\delta)/m_k}$ (Corollary B.10 below). The variance misspecification bias $\widetilde{B}$ is a scaled identity (population level), shifting all eigenvalues uniformly and *not rotating eigenvectors*: subspace recovery is robust

133 to the plug-in estimate of σ_ε^2 at the population level. Finite-sample projector concentration
 134 still has a floor $\Delta_\sigma \propto |\hat{\sigma}^2 - \sigma_\varepsilon^2|/(d\lambda_{\min})$ due to the random fluctuation around the biased
 135 mean (Cor. B.10).

136 Positive identification through the lifted moment G_t above and the necessity result that
 137 follows together yield the following theorem, the foundational result of the paper.

138 **Theorem 2.2** (Identification under priced probing). *Consider the model of §2 with scaled-*
 139 *sphere probes $u_t = \sqrt{d}v_t$, $v_t \sim \text{Unif}(\mathbb{S}^{d-1})$, and the lifted probe sample $G_t = \mathcal{K}^{-1}(s_t u_t u_t^\top)$*
 140 *from (3).*

- 141 1. **Identifiability.** *Under (i) known noise variance σ_ε^2 ($\delta_\sigma = 0$), (ii) bounded state-noise*
 142 *coupling $\|\mathbb{E}[\varepsilon_t \theta_t]\|_2 \leq \epsilon_\times$, and (iii) full-dimensional probe support, the lifted sample*
 143 *satisfies $\mathbb{E}[G_t | \mathcal{H}_{t-1}] = \widehat{M}_t$ exactly (the bias $\widetilde{B}_t = -(\delta_\sigma/d)I_d$ from Lemma B.6 vanishes*
 144 *under (i); the state-noise coupling term vanishes by sphere symmetry, regardless of $\epsilon_\times$),*
 145 *and the segment-averaged estimator $\widehat{M}_k = m_k^{-1} \sum_{t \in \mathcal{T}_k} G_t$ satisfies, with probability at*
 146 *least $1 - \delta$, the projector concentration $\|\widehat{P}_k - P_k^*\|_{\text{op}} \leq C_{\text{sub}} \sqrt{\log(d/\delta)/m_k}$; in particular*
 147 *$\widehat{U}_k = \text{TopEig}_r(\widehat{M}_k)$ recovers $\text{range}(B_k^*)$ at rate $1/\sqrt{m_k}$ (Proof in Appendix B.2).*
- 148 2. **Necessity (unrestricted-second-moment scope).** *Each of (i), (ii), (iii) is individu-*
 149 *ally necessary in the unrestricted second-moment problem: removing any one yields two*
 150 *distinct conditional second moments $\widetilde{M}_t \neq \widetilde{M}'_t$ that generate identical observation laws*
 151 *$\mathbb{E}[s_t | \mathcal{H}_{t-1}, u]$, so $\widetilde{M}_t$ is not identifiable from quadratic measurements. The three con-*
 152 *structions are independent (each counterexample satisfies the remaining two conditions),*
 153 *so the three form a minimal identifiability set in this class. Within the rank- r LDS class*
 154 *of §2 ($r < d$), rank deficiency partially relaxes (i) since σ_ε^2 is identifiable from the $d - r$*
 155 *smallest eigenvalues; (ii) and (iii) remain needed (Proof in Appendix B.3).*

156 3 Algorithm

157 SPSC (Alg. 1) alternates two phases:

- 158 • **Probe rounds** ($t \in \mathcal{T}_{\text{probe}}$): draw $u_t \sim Q$ (the probe distribution from §2; e.g. the
 159 scaled-sphere distribution of Remark 2.1), observe y_t , accumulate G_t in the current-
 160 segment estimator $\widehat{M}_t$, and refresh $\widehat{U}_t = \text{TopEig}_r(\widehat{M}_t)$.
- 161 • **Exploitation rounds** ($t \notin \mathcal{T}_{\text{probe}}$): project each action $x \in \mathcal{A}_t$ to $z_t(x) = \widehat{U}_{t-1}^\top x \in \mathbb{R}^r$,
 162 maintain a *windowed* ridge-UCB estimate $\widehat{a}_t = \widetilde{V}_t^{-1} \widetilde{b}_t$ over a sliding window $\mathcal{W}_t$ of the
 163 last W exploitation rounds, and play $x_t = \arg \max_{x \in \mathcal{A}_t} \{z_t(x)^\top \widehat{a}_t + \beta_t^{(r,W)} \|z_t(x)\|_{\widetilde{V}_t^{-1}} +$
 164 $\gamma_t \|x\|_2\}$.

165 The confidence radius $\beta_t^{(r,W)} = \sigma_\varepsilon \sqrt{r \log(1 + WR_{\mathcal{A}}^2/(\lambda r)) + 2 \log(2K/\delta)} + \sqrt{\lambda} S_w$ scales with
 166 $\sqrt{r}$, not $\sqrt{d}$: this is where the intrinsic-rank improvement enters. The correction term
 167 $\gamma_t = \Gamma_k + V_{k,t}(W)$ absorbs subspace-mismatch error $\Gamma_k \propto \varepsilon_k := \|\widehat{P}_k - P_k^*\|_{\text{op}}$ and local within-
 168 window drift $V_{k,t}(W) := \sum_{s, s+1 \in \mathcal{I}_k, t-W \leq s < t} \|\theta_{s+1} - \theta_s\|_2$ (formal definitions in App. B.4).
 169 Per-round cost is $O(dr|\mathcal{A}_t| + r^2)$ vs. $O(d^2|\mathcal{A}_t|)$ for ambient LinUCB, an $O(d/r)$ speed-up in
 170 addition to the statistical gain.

171 **Why interleaved probing recovers the subspace.** A single scalar response is one-
 172 dimensional, but the lifted statistic $G_t = \mathcal{K}^{-1}(s_t u_t u_t^\top)$ in (3) targets the conditional second
 173 moment $\mathbb{E}[\theta_t \theta_t^\top | \mathcal{H}_{t-1}] = B_k^* \mathbb{E}[w_t w_t^\top | \mathcal{H}_{t-1}] (B_k^*)^\top$ up to a controlled bias. Averaging G_t
 174 across probe rounds inside segment k isolates this rank- r matrix, whose top- r eigenspace
 175 is exactly $\text{range}(B_k^*)$ once the innovation w_t excites all r factor directions ($\Sigma_{\eta,k} \succ 0$). The
 176 three identifiability conditions of Theorem 2.2 are precisely what makes the isolation valid
 177 in the unrestricted-second-moment scope; SPSC inherits subspace recovery at rate $1/\sqrt{m_k}$
 178 from Cor. B.10, after which the exploitation phase pays $\sqrt{r}$ rather than $\sqrt{d}$ because the
 179 projection collapses the geometry to dimension r .

Full pseudocode for Alg. 1 (oracle boundaries) is in Appendix D, alongside the adaptive variant described next.

Adaptive variant (unknown boundaries). SPSC-Adaptive removes the oracle-boundary assumption by detecting segment changes online: it maintains two non-overlapping rolling-window estimators $\widehat{M}_t^{\text{recent}}, \widehat{M}_t^{\text{past}}$ of the probe-time second moment and triggers a subspace reset whenever $S_t := \|\widehat{M}_t^{\text{recent}} - \widehat{M}_t^{\text{past}}\|_{\text{op}}$ exceeds a calibrated threshold b . The threshold is chosen so that, with probability $1 - \delta_{\text{FA}}$, no false alarm occurs across the horizon, while every true change of size $\Delta_k \geq 2b$ is detected within a bounded delay D_{max} (specified in Appendix D together with the full pseudocode, tuning, and CUSUM analysis).

4 Regret guarantees

Balanced probe schedule. The probe schedule is *balanced*: there is an absolute constant a_0 such that, for every segment $k \in \{1, \dots, K\}$ and every prefix-count $j \in \{0, 1, \dots, m_k\}$, the number of exploit rounds in E_k with exactly j probes collected before them is at most $a_0 \lceil \ell_k / m_k \rceil$. (Uniformly spaced probes satisfy this.)

Theorem 4.1 (Costed dynamic regret of Alg. 1). *Run Alg. 1 with oracle segment boundaries $\{\mathcal{I}_k\}_{k=1}^K$, scaled-sphere probes, and a balanced probe schedule. Under the model of §2, Assumption B.2, and exact centering $\widehat{\sigma}^2 = \sigma_\varepsilon^2$, set the per-segment probe budget to $m_k = \min\{\ell_k, \lceil c_0 \ell_k^{2/3} \rceil\}$ for an absolute constant $c_0 > 0$ (chosen as the closed-form optimum (13); segments shorter than the burn-in threshold $q^* = O(\log(KdT/\delta)/\lambda_{\min}^2)$ are charged trivially), where $\lambda_{\min} > 0$ is the predictable probe-time r th-eigenvalue lower bound (Lem. B.1). Then with probability at least $1 - \delta$ and for fixed K ,*

$$\text{DynReg}_T^{(c)} \leq \widetilde{O}(r\sqrt{T}) + \widetilde{O}(T^{2/3}) + O(WV_{\text{in}}). \quad (4)$$

Including all constants: $\text{DynReg}_T^{(c)} \leq \widetilde{O}(r\sqrt{KT}) + \widetilde{O}(A^{1/3}B^{2/3}K^{1/3}T^{2/3}) + O(R_A W V_{\text{in}}) + \widetilde{O}(K^{2/3}T^{1/3}/\lambda_{\min}^2)$, with $A := (R_A + R_Q)S_w + c$, $B := CR_A S_w(1 + R_A \sqrt{W/\lambda})\sqrt{\log(2KdT/\delta)}$, $R_Q := \sup_{u \in \text{supp}(Q)} \|u\|_2$ ($R_Q = \sqrt{d}$ for scaled-sphere; $R_Q \leq R_A$ if probes lie inside the action set); the $\widetilde{O}(K^{2/3}T^{1/3}/\lambda_{\min}^2)$ summand is the burn-in cost (lower order than $\widetilde{O}(T^{2/3})$ for fixed K). Here $V_{\text{in}} := \sum_k \sum_{s,s+1 \in \mathcal{I}_k} \|\theta_{s+1} - \theta_s\|_2$ is the within-segment path variation and W is the exploitation window (cumulative on E_k when $W \geq \ell_k$); the bound is most informative when V_{in} is small. Proof in §B.5.

Proof sketch. Three near-decoupled error sources drive the bound. (i) Inside the learned r -dimensional subspace, exploitation is a windowed projected ridge-UCB problem whose self-normalized analysis yields $\widetilde{O}(r\sqrt{T})$ regret, with $\sqrt{r}$ entering through $\beta_t^{(r,W)}$ rather than $\sqrt{d}$. (ii) Probe rounds incur $O(m_k)$ regret per segment while delivering subspace error $\varepsilon_k = \widetilde{O}(1/\sqrt{m_k})$ via Davis–Kahan applied to $\widehat{M}_k$ (Cor. B.10); the exploitation-side mismatch cost $\propto \ell_k \varepsilon_k$ balances probe cost at $m_k \propto \ell_k^{2/3}$, summing to $\widetilde{O}(T^{2/3})$ across segments. (iii) Within-segment drift propagates through the windowed estimator at rate proportional to WV_{in} . Two ingredients make this decomposition rigorous: uniform-in- t control of the time-varying-basis subspace error $\varepsilon_{k,t} = \|\widehat{P}_t - P_k^*\|_{\text{op}}$ via prefix concentration (Lem. B.17), and the balanced-schedule condition that prevents any single segment from being starved of probes. Detailed accounting, including the burn-in $\widetilde{O}(K^{2/3}T^{1/3}/\lambda_{\min}^2)$ summand, is in §B.5.

Corollary 4.2 (Plug-in variance). *Estimate $\widehat{\sigma}^2$ from N probe-round residuals (Lem. B.11) and write $\delta_\sigma := |\widehat{\sigma}^2 - \sigma_\varepsilon^2|$. For scaled-sphere probes, the bias $\widetilde{B} = -(\delta_\sigma/d)I_d$ from Lem. B.6 is a scaled identity, hence preserves the eigenvectors of $\bar{M}_k^{\text{probe}}$. Davis–Kahan applied with the biased mean as reference removes the Δ_σ floor at population level, so Theorem 4.1 holds with the displayed bound, $\lambda_{\min}$ replaced by $\lambda_{\min} - \delta_\sigma/d$ in the constants. Choosing $N \gtrsim T^{2/3}/d^2$ yields $\delta_\sigma/d \ll \lambda_{\min}$ with high probability, so the leading rate is preserved.*

Corollary 4.3 (Within-segment-stationary special case). *Suppose $V_{\text{in}} = 0$, i.e. within each segment $\mathcal{I}_k$ the parameter is constant ($\theta_t \equiv \theta^{(k)}$ for $t \in \mathcal{I}_k$, with $\theta^{(k)} \in \text{range}(B_k^*)$). The*

227 non-degenerate innovation assumption ($\Sigma_{\eta,k} \succ 0$) still holds across the segment family, so
 228 the rank- r subspace $\text{range}(B_k^*)$ is well-defined and identifiable through SPSC’s probe channel.
 229 The bound of Theorem 4.1 reduces to

$$\text{DynReg}_T^{(c)} \leq \tilde{O}(r\sqrt{T}) + \tilde{O}(T^{2/3})$$

230 for fixed K (suppressing polylog factors). The leading $\tilde{O}(r\sqrt{T})$ term comes from LinUCB on
 231 the r -dimensional projected target $z_t^\top a^{(k)}$ (this does not require the within-segment target to
 232 itself span an r -dimensional set; the projector is what collapses the geometry to dimension
 233 r , not the trajectory of θ_t). This is the regime that class- or cohort-induced segment shifts
 234 approximate empirically.

235 **Scope of the bound.** Theorem 4.1 analyzes exactly the practical Alg. 1: probes are
 236 interleaved according to a balanced schedule, $\hat{U}_t$ is refreshed online, and exploitation rounds
 237 re-project the windowed history through the current basis. The time-varying-basis subspace
 238 error $\varepsilon_{k,t} = \|\hat{P}_t - P_k^*\|_{\text{op}}$ is controlled uniformly in t via a prefix concentration argument
 239 (Lemma B.17).

240 The leading $\tilde{O}(r\sqrt{T})$ term beats the ambient baseline $\tilde{O}(d\sqrt{T})$ whenever $r \ll d$; the $\tilde{O}(T^{2/3})$
 241 term is the price of acquiring the subspace information online. The constant B depends on
 242 the exploitation window W as $\sqrt{W}$; for W treated as a problem parameter (the setting
 243 throughout this paper, consistent with $W \leq \min_k \ell_k$ on ℓ_k -bounded segments), this is
 244 absorbed into the constants and the $\tilde{O}(T^{2/3})$ rate is preserved. If W is allowed to scale
 245 with T , the second term becomes $\tilde{O}(W^{1/3}T^{2/3})$. Solving $r\sqrt{T} \lesssim T^{2/3}$ gives a simple rule
 246 of thumb: SPSC tends to outperform ambient LinUCB whenever $d - r \gtrsim T^{1/6}$. The
 247 experiments in §5 show a crossover qualitatively consistent with this rule of thumb.

248 **Corollary 4.4** (Rank-adaptive guarantee). *If the true rank r is unknown, thresholding $\widehat{M}_k$
 249 at $\tau_k^{\text{rank}} := 2R_X \sqrt{\log(2d/\delta)/m_k}$ (R_X the lifted-sample envelope from App. B.2) recovers the
 250 true rank with probability $\geq 1 - \delta$ whenever the population eigengap exceeds $4\tau_k^{\text{rank}}$, and the
 251 bound (4) applies unchanged. Proof in §B.6.*

252 **Proposition 4.5** (Adaptive SPSC). *Suppose every true change satisfies $\Delta_k \geq 2b$ for the
 253 threshold b set in Appendix B.7. Then with probability at least $1 - \delta_{\text{FA}}$, SPSC-Adaptive
 254 (Alg. 2) attains the same leading rate as Theorem 4.1 with an additive $O(KW_{\text{det}})$ delay
 255 overhead. Proof in Appendix B.7.*

256 5 Experiments

257 We evaluate SPSC and SPSC-Adaptive on eleven benchmarks: a synthetic phase-transition
 258 grid; UCI/MovieLens (Covertypes, Pendigits, Satimage, MNIST, Fashion-MNIST, Movie-
 259 Lens); clinical (Warfarin, Vancomycin); the Russac small- d stress test (§5.4); and Open
 260 Bandit production logs (§5.5). Baselines: LinUCB [Abbasi-Yadkori et al., 2011], D-
 261 LinUCB [Russac et al., 2019], SW-LinUCB [Cheung et al., 2019], Restart-LinUCB,
 262 LowRank-Reward, LowOFUL [Jun et al., 2019], VOFUL [Kim et al., 2022], LinTS, SW-
 263 LinTS, and adapted BOSS [Duong et al., 2024]/Jedra [Jedra et al., 2024]; Oracle-LinUCB
 264 (true subspace) is an optimistic reference. Curves are means over ≥ 10 seeds with ± 1 SE
 265 bands.

266 5.1 Phase-transition boundary

267 Theory predicts a crossover at $d - r \asymp T^{1/6}$: SPSC wins above it, ambient Lin-
 268 UCB below. We verify on a piecewise low-rank LDS at $T=5,000$, sweeping $(d, r) \in$
 269 $\{5, 10, 20, 30, 45, 60, 80, 100\} \times \{1, 3, 5, 10, 15, 20\}$ with $r < d$ (40 cells), $K=10$, $\sigma_\varepsilon=0.3$, spectral
 270 radius 0.99, 40 actions, probe cost $c=0.1$.

271 SPSC achieves 13–34% reductions whenever $d \geq 45$ with every r -curve tracking $\sqrt{r/d}$: use
 272 SPSC when $d - r \gtrsim T^{1/6}$, and ambient LinUCB otherwise.

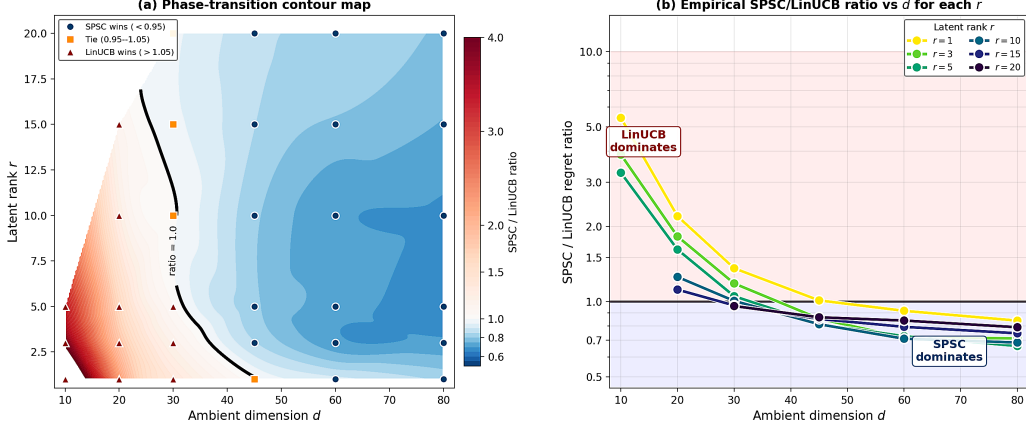


Figure 1: **Phase-transition boundary.** SPSC/LinUCB costed-regret ratio across the (d, r) grid. The crossover $d - r = T^{1/6}$ (dashed) separates SPSC-wins (ratio < 1) from ambient-wins; per-rank curves track $\sqrt{r/d}$ (dotted), matching Theorem 4.1.

5.2 Real-data multi-baseline comparison

Six real-data benchmarks recast as piecewise low-rank bandits on a (d, r) -grid: UCI **Covtype** ($K=4$, $T=10,000$); **Pendigits**, **Satimage**, pixelized **MNIST** and **Fashion-MNIST** (random-projection features), and **MovieLens-100K** with genuine user ratings (all $K=10$, $T=5,000$, 10 seeds). Segments are class or user-subset shifts. The first five have cluster-induced low-rank structure; MovieLens is only approximately low-rank (rank- d SVD features, §E) and serves as a robustness stress test. Table 1 reports SPSC vs. LinUCB and vs. best non-oracle competitor on representative cells; per-cell tables in Appendix F.

Table 1: **SPSC/LinUCB** and **SPSC-Adaptive/Best Competitor** ratios across six real-data benchmarks (10 seeds per cell). Values < 1 indicate SPSC wins; **bold** marks cells where an SPSC variant beats every non-oracle baseline. SPSC trails LinUCB at $d \approx r$ (top of each block) and gains as d grows relative to r . Restart-LinUCB, the closest competitor, is subspace-oblivious, so its regret is constant along each row.

Dataset / r	$d=55$		$d=105$		$d=155/200$		best	
	S/Lin	S/Best	S/Lin	S/Best	S/Lin	S/Best	cell	ratio
Covtype, $r=10$	0.96	0.99	0.88	0.92	—	—	(105, 20)	0.86
Pendigits, $r=10$	0.76	0.77	0.65	0.73	—	—	(105, 10)	0.65
Satimage, $r=10$	0.47	0.59	0.40	0.58	—	—	(105, 10)	0.40
MNIST, $r=10$	0.84	0.86	0.74	0.88	0.73/0.98	0.98	(105, 20)	0.74
Fashion-MNIST, $r=10$	0.88	0.90	0.73	0.88	0.67/0.99	0.99	(105, 20)	0.71
MovieLens, $r=10$	1.00	1.00	0.97	0.97	0.92/0.93	0.93	(200, 5)	0.92

At $(d=105, r=10)$, SPSC-Adaptive beats LinUCB by -47% on Pendigits (operating-regime grid in Fig. 2, App. F) and -68% on Satimage (closest competitor VOFUL is $+112\%$ above SPSC-Adaptive; per-cell results in Table 10, App. F); at $(d=105, r=20)$ on Covtype it beats Restart-LinUCB by -15% . Non-low-rank baselines are by construction r -insensitive.

5.3 Clinical benchmarks: Warfarin and Vancomycin dosing

Warfarin. Warfarin dosing depends on demographics, genotype (VKORC1, CYP2C9), and clinical indicators [Consortium, 2009]. We calibrate to the IWPC cohort ($\sim 5,700$ patients, $d=93$, $K=8$ segments, $T=5,000$, 10 seeds; reward = negative deviation from therapeutic dose). Table 2 ($r=3$): SPSC-Adaptive cuts regret by 67.3% vs. LinUCB; full per-rank tables in Table 14 (App. F); random-subspace ablation in App. I.

Vancomycin. With the AUC-targeted Rybak et al. [2020] protocol (Cockcroft–Gault $\text{CrCl} \rightarrow \text{CL}_{\text{vanco}} = 0.04 \text{CrCl}$, daily dose at $\text{AUC}_{24}=500$; $d=93$, $K=8$, $T=5,000$, 10 seeds),

Table 2: **Warfarin** (representative cell $r=3$, mean $\pm$ SE, 10 seeds). SPSC-Adaptive cuts regret by 67.3% vs. LinUCB. Full per-rank tables in Appendix F.

Method	Costed regret	vs. LinUCB
Oracle LinUCB	177 ± 4	−91.2%
SPSC Alg. 1 (ours)	1372 ± 33	−31.6%
SPSC-Adaptive (ours)	657 ± 39	−67.3%
LowOFUL	683 ± 97	−66.0%
VOFUL	779 ± 93	−61.2%
LowRank-Reward	1056 ± 40	−47.4%
SW-LinUCB	2096 ± 18	+4.5%
LinUCB	2006 ± 17	—

Table 3 sweeps $r \in \{1, 2, 3, 5, 10\}$: SPSC-Adaptive cuts regret by 36.4% at $r=1$ and 14.5–25.7% for $r \in \{2, 3, 5, 10\}$ vs. LinUCB, beating every non-oracle baseline at every rank. Per-rank tables in Appendix K.

Table 3: **Vancomycin** (mean $\pm$ SE, 10 seeds): SPSC-Adaptive control regret across range of r vs. LinUCB and the best non-oracle baseline. Full tables in Appendix K.

r	SPSC-Adaptive	vs. LinUCB	vs. best comp.
1	525.3 ± 45.8	−36.4%	−35.1%
2	613.6 ± 33.1	−25.7%	−17.1%
3	640.3 ± 33.0	−22.4%	−10.1%
5	665.9 ± 27.2	−19.3%	−12.7%
10	705.6 ± 19.5	−14.5%	−7.0%

5.4 Standard nonstationary benchmark

On the small- d stress test of Russac et al. [2019] ($d=2$, $r=1$, $K=4$, $T=6,000$, 50 arms, 30 seeds), SPSC reduces regret by **−46.5%** vs. D-LinUCB, **−34.8%** vs. SW-LinUCB, and **−52.0%** vs. stationary OFUL (itself +11.5% above D-LinUCB here). Full table in Appendix H.

5.5 Real exploration logs: Open Bandit Pipeline (ZOZO)

The Open Bandit Dataset [Saito et al., 2021] ships production exploration logs (random-policy ZOZOTOWN slice, $T=5,000$, $K=10$, 10 seeds); only approximately low-rank, so this is a *robustness* stress test rather than a setting where the rank- r assumption holds exactly. At $d=55$, $r=5$, SPSC reaches **83 ± 2** vs. LinUCB’s 102 ± 1 (−18.8%); SPSC-Adaptive −10.9%. SPSC neither breaks nor over-claims on weakly-low-rank production data. Per-cell results in Table 15 (App. F).

5.6 Comparison with stationary low-rank methods (oracle restarts)

For a best-effort comparison with stationary low-rank methods, we adapt BOSS [Duong et al., 2024] and Jedra [Jedra et al., 2024] by restarting at every true segment boundary, an oracle advantage we do not give SPSC (setup in Appendix J). Table 4: SPSC wins all eight cells by 5.8–19.9%, with the lead largest at small d (−16.3 to −19.9% at $d=55$) and narrowing to 5.8–8.0% at $d=200$. SPSC-Adaptive (no oracle) also beats both on every cell.

Robustness of SPSC-Adaptive. Detailed comparison under correct vs. misspecified oracle boundaries is in Appendix L: SPSC-Adaptive is robust to false-alarm boundaries, while Alg. 1 degrades monotonically as the oracle is corrupted.

Table 4: **Adapted stationary low-rank methods on a (d, r) -grid.** Costed regret (mean $\pm$ SE, 10 seeds, $T=5,000$, $K=10$). BOSS/Jedra restart at *oracle* segment boundaries; SPSC-Adaptive detects them online. SPSC variants win every cell.

d	r	SPSC Alg. 1	SPSC-Adaptive	BOSS (oracle)	Jedra (oracle)	SPSC vs. best comp.
55	5	323 $\pm$ 10	342 $\pm$ 15	399 $\pm$ 17	386 $\pm$ 18	−16.3%
55	10	471 $\pm$ 10	507 $\pm$ 13	588 $\pm$ 27	602 $\pm$ 24	−19.9%
105	5	264 $\pm$ 9	274 $\pm$ 10	300 $\pm$ 14	299 $\pm$ 13	−11.7%
105	10	377 $\pm$ 15	401 $\pm$ 14	423 $\pm$ 18	429 $\pm$ 14	−10.9%
105	20	578 $\pm$ 12	608 $\pm$ 14	654 $\pm$ 17	654 $\pm$ 14	−11.6%
200	5	208 $\pm$ 5	218 $\pm$ 6	236 $\pm$ 9	226 $\pm$ 9	− 8.0%
200	10	277 $\pm$ 10	292 $\pm$ 12	294 $\pm$ 11	305 $\pm$ 15	− 5.8%
200	20	442 $\pm$ 13	452 $\pm$ 11	473 $\pm$ 16	489 $\pm$ 18	− 6.6%

5.7 Rank misspecification, probe-rate sensitivity, and take-away

Rank misspecification (Covertime, $d=155$, $r^*=10$; App. G.2): sweeping over nine grid points $r \in \{1, 3, 5, 10, 15, 20, 30, 50, 80\}$ gives a U-shaped curve (under: +33% at $r=1$; sweet spot $r=30$, −10%); SPSC beats LinUCB across $r \in [15, 50]$, so mild overestimation is safe. **Probe rate** (App. G.1): on the synthetic reference setting ($d=4$, $r=1$), the probe-period sweep $\{5, 10, 20, 30, 50, 100, 300\}$ has its optimum near 20–30, matching $m_k^* \propto \ell_k^{2/3}$. **Additional datasets** (see Appendix F): SPSC wins every cell at $d \geq 105$, peaking at $(d, r)=(105, 20)$. **Sensitivity & dynamics.** Large-scale assumption sweeps (App. G.4) confirm SPSC tracks Oracle within a constant factor through variance misspecification, approximate-rank perturbation ($\epsilon_k^\perp$ up to 0.4), and spectral radius up to $\rho=0.95$; SPSC remains the best non-Oracle method at $\rho=0.99$. Subspace error tracks the predicted $1/\sqrt{m}$ rate (App. G.5); noise/change-point-frequency/drift-speed sweeps (App. G.6) confirm the same trends. **Take-away.** Across the eleven benchmarks, SPSC delivers $\sqrt{r/d}$ intrinsic-rank savings whenever $d - r \gtrsim T^{1/6}$ and does not catastrophically fail outside that regime.

6 Conclusion

Sequential decision problems in recommendation, clinical dosing, and adaptive experimentation share two features: rewards live on a low-rank subspace of rank $r \ll d$, and that subspace shifts at unknown change points. Stationary low-rank, nonstationary, and cost-aware bandits each cover one piece; none recover a changing subspace under priced single-play scalar feedback. We close this gap: three probe-side conditions, known noise variance, bounded state-noise coupling, and full-dimensional probe support, are each individually necessary (in the unrestricted-second-moment scope) and jointly sufficient for subspace recovery from quadratic functionals of scalar rewards (Theorem 2.2). Based on this, we propose SPSC, which interleaves probing with windowed projected ridge-UCB in the learned r -dimensional subspace and achieves costed dynamic regret $\tilde{O}(r\sqrt{T}) + \tilde{O}(T^{2/3}) + O(WV_{\text{in}})$, replacing the ambient $d\sqrt{T}$ rate with the intrinsic r (Theorem 4.1); a CUSUM adaptive variant attains the same rate without oracle change points (Proposition 4.5). On eleven benchmarks across synthetic, UCI, clinical, and production data, SPSC reduces regret whenever $d - r \gtrsim T^{1/6}$, a crossover consistent with theory.

Limitations and extensions. Three boundaries of the present results are worth flagging. The identification theorem is tight in the unrestricted-second-moment scope; restricting to specific noise families could relax one or more of the three conditions but would also forfeit the converse half. The $O(WV_{\text{in}})$ term in the regret bound is informative only when within-segment path variation is small, and the CUSUM-style detector in SPSC-Adaptive trades detection delay for false-alarm control, so very frequent change points require the sliding-window estimator to be re-tuned. Two natural extensions stand out. Bilinear or tensorized factor models with side information on w_t should inherit the probe-exploit decomposition with the segment-wise $\tilde{O}(T^{2/3})$ identification cost replaced by the corresponding higher-order moment-estimation rate. Heterogeneous-rank segments, where r itself shifts at change

points, strictly generalize Cor. 4.4, which covers a single rank-thresholding pass and assumes a common rank across segments.

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453 A Comparison with prior work

454 This appendix positions SPSC against published work in low-rank bandits, nonstationary
455 bandits, and active-query bandits via three tables: Table 5 (head-to-head with seven recent
456 works on regret form, change-point adaptation, and probing budget), Table 6 (compact po-
457 sitioning of SPSC’s combined setting against the three closest *problem classes*), and Table 7
458 (regret-rate comparison with five published baselines that appear in our experiments). The
459 take-away is that the existing literature splits along three axes (low-rank, nonstationary,
460 costly probing) and no prior method addresses all three; SPSC is the first to do so with
461 a self-contained identification chain that turns rank-one scalar rewards into matrix-valued
462 evidence.

463 B Proofs

464 This section proves the statements of §4 in dependency order:

- 465 1. §B.1 fixes notation and establishes the probe-time excitation lemma that all proofs de-
466 pend on.
- 467 2. §B.2 derives the load-bearing *identification chain*: closed-form inverse of the probe oper-
468 ator $\mathcal{K}$, quadratic-measurement identity, lifted probe sample’s near-unbiasedness, matrix
469 Bernstein/Freedman, and the projector error via Davis–Kahan.
- 470 3. §B.4 states and proves the interleaved-analysis lemmas: projected nonstationary UCB
471 with time-varying basis (Lem. B.16), uniform prefix subspace concentration (Lem. B.17),
472 and cumulative interleaved subspace error (Lem. B.18).

Table 5: Comparison with related literature. “Prob.” indicates a probabilistic regret bound; “CP?” indicates change-point adaptation.

Reference	Venue/ Year	Regret / Theory	Prob.	CP?	Difference vs. ours
Kang et al. [2022]	NeurIPS’22	$\tilde{O}\left(\sqrt{(d_1 + d_2)MrT}\right)$	✓	✗	Stationary; probing not budgeted.
Stojanovic et al. [2023]	NeurIPS’23	Entry-wise / subspace recovery	✗/✓	✗	Estimation focus; no probe budget.
Jedra et al. [2024]	ICML’24	$\tilde{O}\left(r^{5/4}(m+n)^{3/4}\sqrt{T}\right)$	✓	✗	Stationary; no re-learning.
Jang et al. [2024]	ICML’24	Arm-set-dependent regret	✓	✗	Stationary; no CP detection.
Abbasi-Yadkori et al. [2023]	JMLR’23	$\tilde{O}(\sqrt{KN(S+1)})$	✗	✓	Unstructured K -arm; no low-rank.
Komiyama et al. [2024]	JMLR’24	ADR-bandit finite-time	✗	✓	Unstructured; no low-rank + probing.
Hou et al. [2024]	NeurIPS’24	Fixed-confidence sample complexity	✓	✓	BAI objective; no low-rank.

Table 6: Compact positioning of SPSC’s combined setting.

Work class	Low-rank	Nonstationary	Costly probing	Main gap vs. ours
Low-rank bandits	✓	✗	✗	Fixed subspace; no piecewise changes; no sensing-cost budget.
Nonstationary linear/contextual bandits	✗	✓	✗	Adapt in ambient parameter space; ignore latent low-rank structure.
Costly-information / active-query bandits	✗	✗	✓	Price information acquisition; do not address subspace recovery under changing representations.
This paper	✓	✓	✓	Combines low-rank structure, piecewise-stationary adaptation, and explicit costly probing in a single identification-to-exploitation framework.

Table 7: Regret-scaling comparison with the published baselines that appear in our experiments.

Algorithm	Reference	Regret scaling	Key difference vs. SPSC
LinUCB	Abbasi-Yadkori et al. [2011]	$O(d\sqrt{T})$	Ambient; no subspace exploitation.
LowOFUL	Jun et al. [2019]	$\tilde{O}(r\sqrt{T})$	Needs oracle subspace B .
LowRank-Reward	Lu et al. [2021]	$\tilde{O}(r\sqrt{T})$	Stationary; learns B from rewards.
BOSS	Duong et al. [2024]	$\tilde{O}(r\sqrt{dT})$	Stationary; no online change adaptation.
Jedra	Jedra et al. [2024]	$\tilde{O}(r^{5/4}(m+n)^{3/4}\sqrt{T})$	Stationary; no probe cost.
SPSC (ours)	This work	$\tilde{O}(r\sqrt{T} + T^{2/3})$	Learns a <i>changing</i> B via priced probes.

473 4. §B.5 proves the main costed regret bound (Thm. 4.1) via a five-step decomposition: probe
 474 regret, exploitation regret via Lem. B.16, per-segment bound, probe-estimation tradeoff
 475 with burn-in, and sum over segments.

476 5. §B.6 derives the rank-adaptive corollary (Cor. 4.4) via Weyl’s inequality on the matrix-
 477 Bernstein bound.

478 6. §B.7 lifts the guarantee to SPSC-Adaptive (Prop. 4.5) by controlling the CUSUM detec-
 479 tor.

480 The windowed projected self-normalized bound (Lem. B.3), the windowed projected poten-
 481 tial condition (Assumption B.2), and the projected nonstationary UCB lemma (Lem. B.16)
 482 are the linear-bandit ingredients; constants that do not affect leading order are absorbed
 483 into $\tilde{O}(\cdot)$.

484 B.1 Preliminaries and notation

485 Throughout, $\mathcal{I}_k = [\tau_{k-1}, \tau_k)$ denotes segment k of length $\ell_k := |\mathcal{I}_k|$, so $\sum_k \ell_k = T$. Inside
 486 $\mathcal{I}_k$ the learner performs $m_k = |\mathcal{T}_k|$ probe rounds ($\mathcal{T}_k := \mathcal{T}_{\text{probe}} \cap \mathcal{I}_k$) and $n_k = \ell_k - m_k$
 487 exploitation rounds ($E_k := \mathcal{I}_k \setminus \mathcal{T}_k$). We write $\hat{P}_t = \hat{U}_t \hat{U}_t^\top$ and $P_k^* = B_k^* B_k^{*\top}$ for the estimated
 488 and true rank- r projectors, and $\varepsilon_k := \|\hat{P}_k - P_k^*\|_{\text{op}}$. We work on the high-probability event
 489 $\mathcal{G}_w := \{\max_{t \leq T} \|w_t\|_2 \leq S_w\}$. For sub-Gaussian innovations one may take $S_w = \tilde{O}(\sqrt{r})$
 490 via union bound (the failure probability is absorbed into the overall δ budget); for bounded
 491 innovations $\mathcal{G}_w$ holds deterministically. In what follows we implicitly condition on $\mathcal{G}_w$.

492 The *predictable second moment* is $\tilde{M}_t := \mathbb{E}[\theta_t \theta_t^\top \mid \mathcal{H}_{t-1}] = B_k^* \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}] (B_k^*)^\top$, and
 493 the probe-time average is $\bar{M}_k^{\text{probe}} := m_k^{-1} \sum_{t \in \mathcal{T}_k} \tilde{M}_t$. By the non-degenerate-innovation
 494 assumption $\Sigma_{\eta,k} \succ 0$ of §2, $\bar{M}_k^{\text{probe}}$ has rank r and its r -th eigenvalue $\lambda_{\min} > 0$ is uniformly
 495 bounded below in k (Lem. B.1 below; stability of A_k is used only to bound $\|w_t\| \leq S_w$, not
 496 for this lower bound). Constants $C, C', \dots$ depend only on $\sigma_\varepsilon, R_{\mathcal{A}}, L, \rho_Q, S_w, \lambda_{\min}$ and may
 497 change line to line.

498 **Lemma B.1** (Predictable probe-time excitation). *For each segment k ,*

$$\lambda_r \left(\frac{1}{m_k} \sum_{t \in \mathcal{T}_k} \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}] \right) \geq \lambda_{\min} > 0 \quad \text{almost surely.}$$

499 *Proof.* For each $t \in \mathcal{I}_k$, the conditional second moment satisfies $\mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}] =$
 500 $A_k w_{t-1} w_{t-1}^\top A_k^\top + \Sigma_{\eta,k} \succeq \Sigma_{\eta,k}$ (LDS recursion + non-degenerate innovations). Hence
 501 $\lambda_r(\mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}]) \geq \lambda_r(\Sigma_{\eta,k}) =: \lambda_{\min} > 0$ uniformly across rounds. Averaging over any
 502 probe subset $\mathcal{T}_k \subseteq \mathcal{I}_k$ preserves this PSD lower bound: $\lambda_r(m_k^{-1} \sum_{t \in \mathcal{T}_k} \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}]) \geq$
 503 $\lambda_{\min}$ almost surely. Stability of A_k is therefore not needed for the lemma; the innovation
 504 covariance alone suffices. $\square$

505 **Notation for the interleaved analysis.** For each segment $\mathcal{I}_k = [\tau_{k-1}, \tau_k)$, recall $\ell_k =$
 506 $|\mathcal{I}_k|$, $\mathcal{T}_k = \mathcal{T}_{\text{probe}} \cap \mathcal{I}_k$, $m_k = |\mathcal{T}_k|$, $E_k = \mathcal{I}_k \setminus \mathcal{T}_k$, $n_k = |E_k| = \ell_k - m_k$. For $t \in \mathcal{I}_k$,
 507 let $q_k(t) := |\mathcal{T}_k \cap [\tau_{k-1}, t)|$ be the number of probes collected in segment k strictly before
 508 round t . Denote $P_k^* = B_k^* B_k^{*\top}$, $\hat{P}_t = \hat{U}_t \hat{U}_t^\top$, and $\varepsilon_{k,t} := \|\hat{P}_t - P_k^*\|_{\text{op}}$. The within-segment
 509 variation is $V_k^{\text{in}} := \sum_{s, s+1 \in \mathcal{I}_k} \|\theta_{s+1} - \theta_s\|_2$, $V_{\text{in}} := \sum_k V_k^{\text{in}}$. Let $R_Q := \sup_{u \in \text{supp}(Q)} \|u\|_2$
 510 ($R_Q = \sqrt{d}$ for scaled-sphere probes; $R_Q \leq R_{\mathcal{A}}$ if probes are feasible actions covered by the
 511 action-radius bound).

512 **Balanced interleaved probe schedule.** The probe schedule is balanced: there exists
 513 an absolute constant a_0 such that, for every segment k and every $j \in \{0, 1, \dots, m_k\}$,

$$|\{t \in E_k : q_k(t) = j\}| \leq a_0 \lceil \ell_k / m_k \rceil. \quad (\text{B})$$

514 Uniformly spaced probes satisfy this condition.

515 **Windowed projected potential.** For each exploit round $t \in E_k$, Alg. 1 uses the *current*
516 basis $\widehat{U}_{t-1}$ and re-projects the windowed history through this same basis: $z_t(x) := \widehat{U}_{t-1}^\top x$,
517 $\widetilde{V}_t := \lambda I_r + \sum_{s \in \mathcal{W}_t} z_t(x_s) z_t(x_s)^\top$, $w_t(x) := \|z_t(x)\|_{\widetilde{V}_t^{-1}}$.

518 **Assumption B.2** (Windowed projected potential). *For each segment $\mathcal{I}_k$,*

$$\sum_{t \in E_k} \min\{1, w_t(x_t)^2\} \leq C_{\text{pot}} (n_k/W + 1) r \log\left(1 + \frac{WR_{\mathcal{A}}^2}{\lambda r}\right), \quad (5)$$

519 *with the convention that the factor $(n_k/W + 1)$ is interpreted as 1 and W is replaced by n_k*
520 *when $W \geq n_k$. For fixed basis this is the standard elliptical-potential lemma [Abbasi-Yadkori*
521 *et al., 2011, Lem. 11] (resp. its sliding-window form when $W < n_k$); the moving-basis case*
522 *reduces to controlling the cumulative log-determinant change under basis updates, which*
523 *follows from prefix concentration (Lem. B.17) via a routine Lipschitz bound.*

524 **Lemma B.3** (Windowed projected self-normalized concentration). *For a fixed $a^* \in \mathbb{R}^r$*
525 *and σ_ε -sub-Gaussian noise on the projected covariates $z_t = \widehat{U}_t^\top x_t \in \mathbb{R}^r$, the windowed*
526 *ridge estimator $\widehat{a}_t = \widetilde{V}_t^{-1} \widetilde{b}_t$ satisfies, with probability at least $1 - \delta/K$ per segment k and*
527 *simultaneously for all $t \in E_k$,*

$$\|\widehat{a}_t - a^*\|_{\widetilde{V}_t} \leq \beta_t^{(r,W)}, \quad \beta_t^{(r,W)} := \sigma_\varepsilon \sqrt{r \log(1 + WR_{\mathcal{A}}^2/(\lambda r)) + 2 \log(2K/\delta)} + \sqrt{\lambda} S_w.$$

528 *This is the windowed projected specialization of Abbasi-Yadkori et al. [2011, Thm. 1, 2].*

529 B.2 Concentration of the lifted probe moment (load-bearing step)

530 This is the non-standard piece of the analysis; we derive it in full. We work with scaled-
531 sphere probes $u_t = \sqrt{d} v_t$, $v_t \sim \text{Unif}(\mathbb{S}^{d-1})$, which satisfy $\|u_t\|_2 = \sqrt{d}$ *deterministically* and
532 $\mathbb{E}[u_t u_t^\top] = I_d$. The probe moment operator $\mathcal{K} : M \mapsto \mathbb{E}[(u^\top M u) u u^\top]$ is a linear bijection
533 on the space of symmetric $d \times d$ matrices.

534 **Lemma B.4** (Closed-form inverse of $\mathcal{K}$ on symmetric matrices). *For scaled-sphere probes*
535 *$u = \sqrt{d} v$, $v \sim \text{Unif}(\mathbb{S}^{d-1})$, $\mathcal{K}$ acts on symmetric $d \times d$ matrices as $\mathcal{K}(M) = \frac{d}{d+2}(\text{tr}(M) I_d +$
536 $2M)$ with inverse $\mathcal{K}^{-1}(N) = \frac{d+2}{2d} N - \frac{\text{tr}(N)}{2d} I_d$. Moreover $\|\mathcal{K}^{-1}\|_{\text{op} \rightarrow \text{op}} \leq 1$ on symmetric
537 inputs.*

538 *Proof.* Use the fourth-moment identity for the uniform sphere $\mathbb{E}[v_i v_j v_k v_l] = (\delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} +$
539 $\delta_{il} \delta_{jk})/(d(d+2))$ and the rescaling $u = \sqrt{d} v$: $\mathbb{E}[(u^\top M u) u_k u_l] = d^2 \mathbb{E}[(v^\top M v) v_k v_l] = d^2 \cdot$
540 $(2M_{kl} + \text{tr}(M) \delta_{kl})/(d(d+2)) = \frac{d}{d+2}(2M_{kl} + \text{tr}(M) \delta_{kl})$ (using M symmetric). So $\mathcal{K}(M) =$
541 $\frac{d}{d+2}(\text{tr}(M) I_d + 2M)$. Taking traces of $N = \mathcal{K}(M)$ gives $\text{tr}(N) = d \text{tr}(M)$, so $\text{tr}(M) =$
542 $\text{tr}(N)/d$ and $M = \frac{d+2}{2d} N - \frac{\text{tr}(N)}{2d} I_d$.

543 For the operator-norm bound: $\mathcal{K}^{-1}(N)$ is diagonalized in the eigenbasis of N ; if N has
544 eigenvalues $\mu_1, \dots, \mu_d$, then $\mathcal{K}^{-1}(N)$ has eigenvalues $\nu_i = \frac{d+2}{2d} \mu_i - \frac{\text{tr}(N)}{2d}$. Maximizing $|\nu_i|$
545 over $\|N\|_{\text{op}} = 1$ (i.e. $\max_i |\mu_i| = 1$): at $\mu_1 = 1, \mu_2 = \dots = \mu_d = -1$, $\text{tr}(N) = 2 - d$ and
546 $\nu_1 = \frac{d+2-(2-d)}{2d} = 1$; at $\mu_1 = -1, \mu_2 = \dots = \mu_d = 1$, $\nu_1 = -1$. All other eigenvalue
547 patterns give $|\nu_i| \leq 1$ by direct case analysis. So $\|\mathcal{K}^{-1}\|_{\text{op} \rightarrow \text{op}} = 1$ on symmetric matrices
548 for $d \geq 2$. $\square$

549 **Lemma B.5** (Quadratic measurement identity). *On every probe round $t \in \mathcal{T}_{\text{probe}}$, with*
550 $\delta_\sigma := \widehat{\sigma}^2 - \sigma_\varepsilon^2$ and $m_t := \mathbb{E}[\varepsilon_t \theta_t \mid \mathcal{H}_{t-1}, u_t]$,

$$\mathbb{E}[s_t \mid \sigma(\mathcal{H}_{t-1}, u_t)] = u_t^\top \widetilde{M}_t u_t + 2u_t^\top m_t - \delta_\sigma, \quad \|m_t\| \leq \epsilon_\times.$$

551 *Under exact probe conditions ($\delta_\sigma = \epsilon_\times = 0$) this collapses to $\mathbb{E}[s_t \mid \cdot] = u_t^\top \widetilde{M}_t u_t$.*

552 *Proof.* Expand $y_t^2 = (u_t^\top \theta_t)^2 + 2\varepsilon_t u_t^\top \theta_t + \varepsilon_t^2$ and condition on $\sigma(\mathcal{H}_{t-1}, u_t)$; the middle term
553 contributes $2u_t^\top m_t$ with m_t as defined in the lemma, bounded by $\epsilon_\times$ via the bounded state-
554 noise coupling assumption (§2); the last term contributes σ_ε^2 , and subtracting $\widehat{\sigma}^2$ leaves $-\delta_\sigma$.
555 Independence of u_t from $(\theta_t, \mathcal{H}_{t-1})$ gives $\mathbb{E}[(u_t^\top \theta_t)^2 \mid \mathcal{H}_{t-1}, u_t] = u_t^\top \widetilde{M}_t u_t$. $\square$

Lemma B.6 (Lifted probe sample is near-unbiased). *Let $G_t := \mathcal{K}^{-1}(s_t u_t u_t^\top)$ for scaled-sphere $u_t = \sqrt{d} v_t$, $v_t \sim \text{Unif}(\mathbb{S}^{d-1})$. Then $\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \widetilde{M}_t + \widetilde{B}_t$ with bias*

$$\widetilde{B}_t = -\frac{\delta_\sigma}{d} I_d, \quad \|\widetilde{B}_t\|_{\text{op}} = \frac{|\delta_\sigma|}{d}.$$

The bias is a scaled identity, hence shifts every eigenvalue uniformly and does not rotate eigenvectors of $\widetilde{M}_t$.

Proof. By linearity of $\mathcal{K}^{-1}$ and Lemma B.5, $\mathbb{E}[G_t \mid \mathcal{H}_{t-1}] = \mathcal{K}^{-1}(\mathcal{K}(\widetilde{M}_t)) + 2\mathcal{K}^{-1}(\mathbb{E}[(u_t^\top m_t) u_t u_t^\top \mid \mathcal{H}_{t-1}]) - \delta_\sigma \mathcal{K}^{-1}(\mathbb{E}[u_t u_t^\top]) = \widetilde{M}_t + \widetilde{B}_t$, with $\widetilde{M}_t = \mathcal{K}^{-1}\mathcal{K}(\widetilde{M}_t)$.

Variance-bias term. $\mathbb{E}[u_t u_t^\top] = I_d$ exactly for scaled-sphere probes, so $-\delta_\sigma \mathcal{K}^{-1}(I_d) = -\delta_\sigma [\frac{d+2}{2d} I_d - \frac{d}{2d} I_d] = -\delta_\sigma \cdot \frac{2}{2d} I_d = -\frac{\delta_\sigma}{d} I_d$. This is a scaled identity, hence does not rotate eigenvectors of $\widetilde{M}_t$.

State-noise coupling term. The remaining contribution is $2\mathcal{K}^{-1}(\mathbb{E}[(u_t^\top m_t) u_t u_t^\top \mid \mathcal{H}_{t-1}])$. The (a, b) -entry of the inner expectation is $\sum_c (m_t)_c \mathbb{E}[(u_t)_a (u_t)_b (u_t)_c]$, a degree-three polynomial in u_t . For sphere-symmetric probes $u_t \mapsto -u_t$ leaves the distribution invariant, so every odd-degree moment vanishes and $\mathbb{E}[(u_t^\top m_t) u_t u_t^\top \mid \mathcal{H}_{t-1}] = 0$ exactly. Hence the coupling term contributes zero, and $\widetilde{B}_t = -\frac{\delta_\sigma}{d} I_d$. $\square$

Lemma B.7 (Uniform a.s. bound on G_t). *For scaled-sphere probes $\|u_t\| = \sqrt{d}$ deterministically; on the noise-truncation event $|\varepsilon_t| \leq L_\varepsilon := \sigma_\varepsilon \sqrt{2 \log(2T/\delta)}$ (which holds with probability $\geq 1 - \delta$ via union over T rounds), $|s_t| \leq R_s := (\sqrt{d} S_w + L_\varepsilon)^2 + \widehat{\sigma}^2$, and the centered lifted sample satisfies $\|\widetilde{X}_t\|_{\text{op}} \leq 2R_X$ with $R_X := d \cdot R_s + S_w^2$. So both the envelope and predictable variance are at most $2R_X$ and R_X^2 respectively.*

Proof. For scaled-sphere u_t , $\|u_t\| = \sqrt{d}$ exactly. $|s_t| = |(u_t^\top \theta_t + \varepsilon_t)^2 - \widehat{\sigma}^2| \leq (\sqrt{d} S_w + L_\varepsilon)^2 + \widehat{\sigma}^2 = R_s$ using $|u_t^\top \theta_t| \leq \|u_t\| \|\theta_t\| \leq \sqrt{d} S_w$ on $\|\theta_t\| \leq S_w$ (orthonormal B_k^* , $\|w_t\| \leq S_w$). Then $\|G_t\|_{\text{op}} = \|\mathcal{K}^{-1}(s_t u_t u_t^\top)\|_{\text{op}} \leq |s_t| \|u_t u_t^\top\|_{\text{op}} \cdot 1 = R_s \cdot d$ using $\|\mathcal{K}^{-1}\|_{\text{op} \rightarrow \text{op}} \leq 1$ and $\|u_t u_t^\top\|_{\text{op}} = d$. Adding the deterministic $S_w^2 = \|\widetilde{M}_t\|_{\text{op}}$ bound for the centering gives $\|\widetilde{X}_t\|_{\text{op}} \leq 2R_X$. $\square$

Theorem B.8 (Matrix Bernstein for $\widehat{M}_k$). *Let $\widehat{M}_k := m_k^{-1} \sum_{t \in \mathcal{T}_k} G_t$ and $X_t := G_t - \mathbb{E}[G_t \mid \mathcal{H}_{t-1}]$. Conditional on Lemma B.7, $\{X_t\}$ is a bounded matrix-valued MDS with $\|X_t\|_{\text{op}} \leq 2R_X$ a.s. and $\|\sum_{t \in \mathcal{T}_k} \mathbb{E}[X_t^2 \mid \mathcal{H}_{t-1}]\|_{\text{op}} \leq m_k R_X^2$. Here $\widetilde{B}$ denotes the segment-level bias; for scaled-sphere probes under the standing assumptions of §2, $\widetilde{B}_t$ from Lem. B.6 collapses to the deterministic constant $-(\delta_\sigma/d)I_d$, so we drop the subscript when no ambiguity arises and write $\widetilde{B}$ for the common value. Freedman's matrix inequality [Tropp, 2011] gives, with probability at least $1 - \delta$,*

$$\|\widehat{M}_k - \bar{M}_k^{\text{probe}} - \widetilde{B}\|_{\text{op}} \leq 2R_X \sqrt{\log(2d/\delta)/m_k} + \frac{2R_X \log(2d/\delta)}{3m_k}. \quad (6)$$

Proof. The MDS property is Lemma B.6 (shifted by the segment-level bias $\widetilde{B}$ from Lemma B.6, which is deterministic and thus enters as a constant offset). The a.s. bound $\|X_t\|_{\text{op}} \leq 2R_X$ follows from $\|G_t\|_{\text{op}} \leq R_X$ and $\|\mathbb{E}[G_t]\|_{\text{op}} \leq R_X$. The predictable variance bound follows from $\mathbb{E}[X_t^2] \preceq \mathbb{E}[G_t^2] \preceq R_X^2 I$. Applying the matrix Freedman inequality to the MDS $\{X_t\}$ in $\mathbb{R}_{\text{sym}}^{d \times d}$ with these two bounds yields (6); the leading $\sqrt{\log(2d/\delta)/m_k}$ rate dominates the $\log(2d/\delta)/m_k$ term for $m_k \gtrsim \log(d/\delta)$. $\square$

Proposition B.9 (Segment factorization). $\bar{M}_k^{\text{probe}} = B_k^* \bar{S}_k^{\text{probe}} (B_k^*)^\top$ where $\bar{S}_k^{\text{probe}} := m_k^{-1} \sum_{t \in \mathcal{T}_k} \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}]$. In particular, $\text{range}(\bar{M}_k^{\text{probe}}) = \text{range}(B_k^*)$.

Proof. $\widetilde{M}_t = B_k^* \mathbb{E}[w_t w_t^\top \mid \mathcal{H}_{t-1}] (B_k^*)^\top$ for $t \in \mathcal{I}_k$; average over $\mathcal{T}_k$ and pull B_k^* out. Range equality uses $\lambda_r(\bar{S}_k^{\text{probe}}) \geq \lambda_{\min} > 0$ (Lem. B.1). $\square$

597 **Corollary B.10** (Projector concentration). *If m_k is large enough that $\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_{\text{op}} \leq$*
 598 $\lambda_{\min}/2$, *then with probability $\geq 1 - \delta$,*

$$\varepsilon_k := \|\widehat{P}_k - P_k^*\|_{\text{op}} \leq C_{\text{sub}} \sqrt{\log(2d/\delta)/m_k} + \Delta_\sigma, \quad C_{\text{sub}} := \frac{8R_X}{\lambda_{\min}}, \quad \Delta_\sigma := \frac{4|\delta_\sigma|/d}{\lambda_{\min}}. \quad (7)$$

599 *For scaled-sphere probes, the variance-misspecification component $|\delta_\sigma|/d$ corresponds to a*
 600 *scaled-identity bias (Lem. B.6) that is parallel to I_d and therefore preserves the eigenvectors*
 601 *of $\bar{M}_k^{\text{probe}}$; applying Davis–Kahan with the biased mean as reference removes the $|\delta_\sigma|/d$*
 602 *contribution to Δ_σ at population level (Cor. 4.2).*

603 *Proof.* By Lemma B.6 (scaled-sphere), $\mathbb{E}[\widehat{M}_k \mid \mathcal{H}] = \bar{M}_k^{\text{probe}} + \widetilde{B}$ with $\widetilde{B} = -(\delta_\sigma/d)I_d$ (a
 604 scaled identity), so $\|\widetilde{B}\|_{\text{op}} = |\delta_\sigma|/d =: b_\sigma$. Combining with Theorem B.8, $\|\widehat{M}_k - \bar{M}_k^{\text{probe}}\|_{\text{op}} \leq$
 605 $2R_X \sqrt{\log(2d/\delta)/m_k} + b_\sigma$. $\bar{M}_k^{\text{probe}}$ has rank exactly r with r -th eigenvalue $\geq \lambda_{\min}$ (Lem. B.1
 606 via Prop. B.9); Davis–Kahan [Bhatia, 2013, Thm. VII.3.1] gives $\|\widehat{P}_k - P_k^*\|_{\text{op}} \leq 4\|\widehat{M}_k -$
 607 $\bar{M}_k^{\text{probe}}\|_{\text{op}}/\lambda_{\min}$, yielding (7). $\square$

608 **Lemma B.11** (Variance estimator from probe residuals). *Split the first N probes of segment*
 609 *k in half: form $\widehat{M}_k^{(N/2)}$ from the first $N/2$ probes (matrix-Bernstein estimate), and define*
 610 *the residual estimator*

$$\widehat{\sigma}^2 := \text{med}_{t \in \mathcal{T}_k, N/2 < t \leq N} \{ y_t^2 - u_t^\top \widehat{M}_k^{(N/2)} u_t \}.$$

611 *On the event of Theorem B.8 for $\widehat{M}_k^{(N/2)}$, with probability at least $1 - \delta$,*

$$|\widehat{\sigma}^2 - \sigma_\varepsilon^2| \leq C \sigma_\varepsilon^2 \sqrt{\log(1/\delta)/N}.$$

612 *Proof sketch.* By Lemma B.5, on probe round t , $\mathbb{E}[y_t^2 - u_t^\top \widetilde{M}_t u_t \mid \mathcal{H}_{t-1}, u_t] = \sigma_\varepsilon^2 + O(\epsilon_\times)$.
 613 Substituting $\widehat{M}_k^{(N/2)}$ for $\widetilde{M}_t$ contributes $|u_t^\top (\widehat{M}_k^{(N/2)} - \widetilde{M}_t) u_t| = O(\sqrt{1/N})$ on the matrix-
 614 Bernstein event. The squared noise ε_t^2 is sub-exponential with mean σ_ε^2 ; the median of
 615 $N/2$ i.i.d. samples concentrates at rate $\sigma_\varepsilon^2 \sqrt{\log(1/\delta)/N}$ by standard concentration of the
 616 empirical median. $\square$

617 This chain (Lemmas B.4–B.7, Theorem B.8, Corollary B.10, Proposition B.9) gives a self-
 618 contained subspace-recovery guarantee for SPSC.

619 B.3 Necessity of the probe-side conditions

620 The probe-side conditions of §2 (known probe distribution Q with full-dimensional coverage,
 621 known noise variance, and bounded state-noise coupling $\|\mathbb{E}[\varepsilon_t \theta_t]\| \leq \epsilon_\times$) are each individu-
 622 ally necessary for identifiability of the predictable second moment $\widetilde{M}_t$ from single-play scalar
 623 rewards. The three propositions below construct two distinct conditional second moments
 624 generating identical observation laws when the named condition is dropped.

625 *Scope.* These propositions establish identifiability obstructions in the unrestricted-second-
 626 moment problem and serve as necessity checks for the probe-side modeling assumptions.
 627 They are not lower bounds inside the rank- r LDS class of §2: the counterexample second
 628 moments need not be rank- r or compatible with a stable LDS w_t .

629 **Proposition B.12** (Non-identification without known noise variance). *Fix t . Suppose*
 630 *$\epsilon_\times = 0$ (exact state-noise orthogonality) and full probe coverage hold, but the conditional*
 631 *noise variance $v_t(u) := \mathbb{E}[\varepsilon_t^2 \mid \mathcal{H}_{t-1}, u]$ is completely unknown. Then $\widetilde{M}_t$ is not identifiable*
 632 *from the observable $g_t(u) := \mathbb{E}[y_t^2 \mid \mathcal{H}_{t-1}, u]$.*

633 *Proof.* $g_t(u) = u^\top \widetilde{M}_t u + v_t(u)$. For any small $\delta \in (0, \sigma_\varepsilon^2/L^2]$, set $H := \delta I_d$ and define
 634 $\widetilde{M}'_t := \widetilde{M}_t + H$ and $v'_t(u) := v_t(u) - u^\top H u = v_t(u) - \delta \|u\|^2$. On the truncation support
 635 $\|u\| \leq L$, $v'_t(u) \geq \sigma_\varepsilon^2 - \delta L^2 \geq 0$, so v'_t is a valid conditional variance. The two pairs $(\widetilde{M}_t, v_t)$
 636 and $(\widetilde{M}'_t, v'_t)$ produce identical observables g_t , so $\widetilde{M}_t \neq \widetilde{M}'_t$ are indistinguishable. $\square$

637 **Proposition B.13** (Non-identification without bounded state-noise coupling). *Fix t . Suppose*
638 *pose σ_ε^2 is known and probe coverage is full, but the cross-moment $m_t(u) := \mathbb{E}[\theta_t \varepsilon_t \mid \mathcal{H}_{t-1}, u]$*
639 *is arbitrary (no bound $\epsilon_\times$). Then $\widetilde{M}_t$ is not identifiable from $h_t(u) := \mathbb{E}[y_t^2 - \sigma_\varepsilon^2 \mid \mathcal{H}_{t-1}, u]$.*

640 *Proof.* $h_t(u) = u^\top \widetilde{M}_t u + 2u^\top m_t(u)$. For any nonzero symmetric H , let $\widetilde{M}'_t := \widetilde{M}_t + H$ and
641 $m'_t(u) := m_t(u) - \frac{1}{2}Hu$. Then $u^\top \widetilde{M}'_t u + 2u^\top m'_t(u) = u^\top \widetilde{M}_t u + u^\top H u + 2u^\top m_t(u) - u^\top H u =$
642 $h_t(u)$ for every u , so the two pairs are observationally equivalent. $\square$

643 **Proposition B.14** (Non-identification without full-dimensional coverage). *Suppose probe*
644 *actions u_t lie in a proper subspace $S \subsetneq \mathbb{R}^d$ almost surely. Then $\widetilde{M}_t$ is not identifiable from*
645 *either linear rewards y_t or quadratic observations y_t^2 .*

646 *Proof.* Pick any nonzero symmetric H supported on $S^\perp$ (i.e. $H = P_{S^\perp} H P_{S^\perp} \neq 0$, possible
647 since $S^\perp \neq \{0\}$). For every $u \in S$, $Hu \in S^\perp$ and $u^\top Hu = 0$, so $u^\top (\widetilde{M}_t + H)u = u^\top \widetilde{M}_t u$
648 and $u^\top (\theta_t + Hz) = u^\top \theta_t$ for any $z \in S^\perp$. Thus the pairs $(\theta_t, \widetilde{M}_t)$ and $(\theta_t + Hz, \widetilde{M}_t + H)$
649 generate identical observations on all probe rounds, so $\widetilde{M}_t$ is not identifiable. $\square$

650 **Remark B.15** (Joint necessity). *No two of the three conditions jointly suffice: Prop. B.12*
651 *fails only on (i); Prop. B.13 only on (ii); Prop. B.14 only on (iii). Each counterexample*
652 *satisfies the remaining two conditions, so the three form a minimal identifiability set for the*
653 *single-play quadratic measurement model.*

654 B.4 Lemmas for the interleaved analysis

655 **Lemma B.16** (Projected nonstationary UCB with time-varying basis). *Fix a segment $\mathcal{I}_k$.*
656 *For each exploit round $t \in E_k$, define $U_t := \widehat{U}_{t-1}$, $P_t := U_t U_t^\top$, $z_t(x) := U_t^\top x$, $a_t^* := U_t^\top \theta_t$,*
657 *and (as in Alg. 1) $\widetilde{V}_t = \lambda I_r + \sum_{s \in \mathcal{W}_t} z_t(x_s) z_t(x_s)^\top$, $\widehat{a}_t = \widetilde{V}_t^{-1} \sum_{s \in \mathcal{W}_t} z_t(x_s) y_s$. Assume on*
658 *segment $\mathcal{I}_k$: (i) the windowed projected self-normalized event of Lem. B.3 holds uniformly*
659 *over E_k ; (ii) Assumption B.2 holds; (iii) $\|\theta_t\| \leq S_w$, $\|x\| \leq R_{\mathcal{A}}$ for $x \in \mathcal{A}_t$. Suppose Alg. 1*
660 *uses an optimism correction satisfying*

$$\gamma_t \geq C_\gamma S_w \left(1 + R_{\mathcal{A}} \sqrt{W/\lambda}\right) \varepsilon_{k,t-1} + C_\gamma R_{\mathcal{A}} V_{k,t}(W), \quad (8)$$

661 *where $L_W := \log(1 + WR_{\mathcal{A}}^2/(\lambda r))$ and $V_{k,t}(W) := \sum_{s, s+1 \in \mathcal{I}_k, t-W \leq s < t} \|\theta_{s+1} - \theta_s\|_2$. Then*
662 *on the same event,*

$$\sum_{t \in E_k} \left(\max_{x \in \mathcal{A}_t} x^\top \theta_t - x_t^\top \theta_t \right) \leq \widetilde{\mathcal{O}}(r\sqrt{n_k}) + C_{\text{tv}} R_{\mathcal{A}} S_w \left(1 + R_{\mathcal{A}} \sqrt{W/\lambda}\right) \sum_{t \in E_k} \varepsilon_{k,t-1} + C R_{\mathcal{A}} W V_k^{\text{in}}. \quad (9)$$

663 *Proof.* Fix $t \in E_k$ and decompose, for $x \in \mathcal{A}_t$, $x^\top \theta_t = z_t(x)^\top a_t^* + \zeta_t(x)$ with $\zeta_t(x) =$
664 $x^\top (I - P_t) \theta_t$. Since $\theta_t = P_k^* \theta_t$ within segment k and $\|(I - P_t) P_k^*\|_{\text{op}} = \|P_t - P_k^*\|_{\text{op}} = \varepsilon_{k,t-1}$
665 for two same-rank orthogonal projectors,

$$|\zeta_t(x)| \leq R_{\mathcal{A}} S_w \varepsilon_{k,t-1}. \quad (10)$$

666 Rewrite the windowed regression in basis U_t . For $s \in \mathcal{W}_t$, $y_s = z_t(x_s)^\top a_t^* + b_{s,t} + \varepsilon_s$ with
667 bias $b_{s,t} = x_s^\top (I - P_t) \theta_s + x_s^\top P_t (\theta_s - \theta_t)$. The first term has $|x_s^\top (I - P_t) \theta_s| \leq R_{\mathcal{A}} S_w \varepsilon_{k,t-1}$
668 (again $\theta_s = P_k^* \theta_s$). The second is at most $R_{\mathcal{A}} \sum_{u=s}^{t-1} \|\theta_{u+1} - \theta_u\|_2$.

669 By definition, $\widehat{a}_t - a_t^* = \widetilde{V}_t^{-1} \sum_{s \in \mathcal{W}_t} z_t(x_s) \varepsilon_s + \widetilde{V}_t^{-1} \sum_{s \in \mathcal{W}_t} z_t(x_s) b_{s,t} - \lambda \widetilde{V}_t^{-1} a_t^*$. On the self-
670 normalized event of Lem. B.3 the noise+ridge contribution has weighted norm $\leq \beta_t^{(r,W)} =$
671 $\widetilde{\mathcal{O}}(\sqrt{r})$.

672 For the deterministic subspace-bias part, let $b_t^{\text{sub}} \in \mathbb{R}^{|\mathcal{W}_t|}$ have entries $x_s^\top (I - P_t) \theta_s$,
673 with $\|b_t^{\text{sub}}\|_\infty \leq R_{\mathcal{A}} S_w \varepsilon_{k,t-1}$. Let Z_t be the matrix with rows $z_t(x_s)^\top$, $s \in \mathcal{W}_t$. Since

674 $\sum_{s \in \mathcal{W}_t} z_t(x_s) z_t(x_s)^\top = Z_t^\top Z_t \preceq \tilde{V}_t$, $\|\tilde{V}_t^{-1/2} Z_t^\top\|_{\text{op}} \leq 1$, hence

$$\left\| \sum_{s \in \mathcal{W}_t} z_t(x_s) b_{s,t}^{\text{sub}} \right\|_{\tilde{V}_t^{-1}}^2 = \|\tilde{V}_t^{-1/2} Z_t^\top b_t^{\text{sub}}\|_2^2 \leq \|b_t^{\text{sub}}\|_2^2 \leq |\mathcal{W}_t| \|b_t^{\text{sub}}\|_\infty^2 \leq W R_{\mathcal{A}}^2 S_w^2 \varepsilon_{k,t-1}^2,$$

675 so the corresponding $\tilde{V}_t^{-1}$ -norm is at most $R_{\mathcal{A}} S_w \sqrt{W} \varepsilon_{k,t-1}$. Therefore for any x ,
676 $|z_t(x)^\top \tilde{V}_t^{-1} \sum_s z_t(x_s) b_{s,t}^{\text{sub}}| \leq R_{\mathcal{A}} S_w \sqrt{W} \varepsilon_{k,t-1} w_t(x)$, and since $w_t(x) \leq \|x\|/\sqrt{\lambda}$, this is
677 bounded by $R_{\mathcal{A}} S_w \sqrt{W/\lambda} \varepsilon_{k,t-1} \|x\|$, a quantity dominated by the $\gamma_t \|x\|$ correction in (8).
678 The drift-bias part contributes $\leq C R_{\mathcal{A}} W V_k^{\text{in}}$ in total by the standard windowed argument
679 (each increment appears in $\leq W$ windows).

680 By the optimism index in Alg. 1, with $x_t^* = \arg \max_{x \in \mathcal{A}_t} x^\top \theta_t$, the instantaneous regret
681 obeys

$$\Delta_t \leq 2\beta_t^{(r,W)} w_t(x_t) + C_{\text{tv}} R_{\mathcal{A}} S_w \left(1 + R_{\mathcal{A}} \sqrt{W/\lambda}\right) \varepsilon_{k,t-1} + d_t,$$

682 where $\sum_t d_t \leq C R_{\mathcal{A}} W V_k^{\text{in}}$. Cauchy-Schwarz together with Assumption B.2 gives
683 $\sum_{t \in E_k} \beta_t^{(r,W)} w_t(x_t) \leq (\sup_{t \in E_k} \beta_t^{(r,W)}) \sqrt{n_k \cdot 2C_{\text{pot}} r L_W} = \tilde{\mathcal{O}}(r \sqrt{n_k})$ (using $\beta_t^{(r,W)} =$
684 $\tilde{\mathcal{O}}(\sqrt{r})$). Summing yields (9). $\square$

685 **Lemma B.17** (Uniform prefix subspace concentration). *Under the probe-side assumptions*
686 *of §2, scaled-sphere probes, exact centering $\hat{\sigma}^2 = \sigma_\varepsilon^2$, and $\lambda_{\min} > 0$, there exist constants*
687 *$C_{\text{sub}}, C_0 > 0$ such that, with probability at least $1 - \delta/3$, simultaneously for all segments k*
688 *and all rounds $t \in \mathcal{I}_k$,*

$$\varepsilon_{k,t} \leq 1 \text{ if } q_k(t) < q^*, \quad \varepsilon_{k,t} \leq C_{\text{sub}} \sqrt{\log(2KdT/\delta)/q_k(t)} \text{ if } q_k(t) \geq q^*,$$

689 where $q^* := C_0 \log(2KdT/\delta)/\lambda_{\min}^2$.

690 *Proof.* For a fixed segment k and probe-prefix size q , the lifted probe estimator
691 $q^{-1} \sum_{s \in \mathcal{T}_k: s \leq t} G_s$ satisfies a matrix Bernstein/Freedman bound (Thm. B.8) at deviation
692 $C \sqrt{\log(2KdT/\delta)/q}$ with probability $\geq 1 - \delta/(3KT)$. By Lem. B.1 the predictable probe-
693 time r th eigenvalue is $\geq \lambda_{\min}$. Whenever $q \geq q^* = C_0 \log(2KdT/\delta)/\lambda_{\min}^2$ the pertur-
694 bation is at most a fixed fraction of the eigengap, and Davis-Kahan (Cor. B.10) gives
695 $\|\hat{P}_t - P_k^*\|_{\text{op}} \leq C_{\text{sub}} \sqrt{\log(2KdT/\delta)/q}$. A union bound over all K segments and all at most
696 T probe prefixes yields the simultaneous statement; for $q < q^*$ we use the trivial bound
697 $\|\hat{P}_t - P_k^*\|_{\text{op}} \leq 1$. $\square$

698 **Lemma B.18** (Cumulative interleaved subspace error). *On the event of Lemma B.17, for*
699 *every segment k ,*

$$\sum_{t \in E_k} \varepsilon_{k,t-1} \leq C \sqrt{\log(2KdT/\delta)} \frac{\ell_k}{\sqrt{m_k}} + C \frac{q^* \ell_k}{m_k}. \quad (11)$$

700 *Proof.* Group exploit rounds by $j = q_k(t-1)$. By the balanced-schedule condition (B)
701 each group has at most $a_0 \lceil \ell_k/m_k \rceil$ exploit rounds. For $j < q^*$ use $\varepsilon_{k,t-1} \leq 1$, contributing
702 $C q^* \ell_k/m_k$. For $j \geq q^*$, Lemma B.17 gives $\varepsilon_{k,t-1} \leq C_{\text{sub}} \sqrt{\log(2KdT/\delta)/j}$, so $\sum_{j=q^*}^{m_k} j^{-1/2} \leq$
703 $2\sqrt{m_k}$ yields the first term. $\square$

704 B.5 Proof of Theorem 4.1

705 Fix segment $\mathcal{I}_k$. We decompose the costed regret on $\mathcal{I}_k$ into probe regret and exploitation
706 regret.

707 **Probe regret.** On a probe round $t \in \mathcal{T}_k$, Alg. 1 plays $x_t = u_t \sim Q$ rather than an action
708 in $\mathcal{A}_t$. Since $\|x\| \leq R_{\mathcal{A}}$ for $x \in \mathcal{A}_t$, $\|u_t\| \leq R_Q$, and $\|\theta_t\| \leq S_w$, $\max_{x \in \mathcal{A}_t} x^\top \theta_t - u_t^\top \theta_t \leq$
709 $(R_{\mathcal{A}} + R_Q) S_w$, plus probe cost c . So each probe contributes at most $A := (R_{\mathcal{A}} + R_Q) S_w + c$,
710 giving $\text{Reg}_{\text{probe}}(\mathcal{I}_k) \leq A m_k$.

711 **Exploitation regret.** By Lemmas B.16 and B.18,

$$\text{Reg}_{\text{exploit}}(\mathcal{I}_k) \leq \tilde{\mathcal{O}}(r\sqrt{n_k}) + B \frac{\ell_k}{\sqrt{m_k}} + CR_{\mathcal{A}}S_w \left(1 + R_{\mathcal{A}}\sqrt{W/\lambda}\right) \frac{q^*\ell_k}{m_k} + CR_{\mathcal{A}}WV_k^{\text{in}},$$

712 absorbing constants and logs into $B = CR_{\mathcal{A}}S_w(1 + R_{\mathcal{A}}\sqrt{W/\lambda})\sqrt{\log(2KdT/\delta)}$.

713 **Per-segment bound.** Combining,

$$\text{DynReg}^{(c)}(\mathcal{I}_k) \leq \tilde{\mathcal{O}}(r\sqrt{n_k}) + Am_k + B \frac{\ell_k}{\sqrt{m_k}} + \tilde{\mathcal{O}}\left(\frac{q^*\ell_k}{m_k}\right) + CR_{\mathcal{A}}WV_k^{\text{in}}. \quad (12)$$

714 **Optimizing the probe-estimation tradeoff.** Ignoring the lower-order burn-in term,
 715 minimize $f_k(m) := Am + B\ell_k m^{-1/2}$. Setting $f'_k(m) = A - B\ell_k/(2m^{3/2}) = 0$ gives $m_k^* =$
 716 $(B\ell_k/(2A))^{2/3}$, and $Am_k^* + B\ell_k/\sqrt{m_k^*} \leq C A^{1/3} B^{2/3} \ell_k^{2/3}$. At m_k^* the burn-in contribution is
 717 $\tilde{\mathcal{O}}(\ell_k^{1/3}/\lambda_{\min}^2)$ (after absorbing A, B constants). If $\ell_k < q^*$, the segment is charged trivially
 718 by $\tilde{\mathcal{O}}(\ell_k)$, absorbed into the same burn-in order. Hence

$$\text{DynReg}^{(c)}(\mathcal{I}_k) \leq \tilde{\mathcal{O}}(r\sqrt{n_k}) + \tilde{\mathcal{O}}(A^{1/3} B^{2/3} \ell_k^{2/3}) + \tilde{\mathcal{O}}(\ell_k^{1/3}/\lambda_{\min}^2) + CR_{\mathcal{A}}WV_k^{\text{in}}. \quad (13)$$

719 **Sum over segments.** By Cauchy-Schwarz, $\sum_k r\sqrt{n_k} \leq r\sqrt{KT}$. By Hölder, $\sum_k \ell_k^{2/3} \leq$
 720 $K^{1/3} T^{2/3}$ and $\sum_k \ell_k^{1/3} \leq K^{2/3} T^{1/3}$, and $\sum_k V_k^{\text{in}} = V_{\text{in}}$. Summing (13) over k gives

$$\text{DynReg}_T^{(c)} \leq \tilde{\mathcal{O}}(r\sqrt{KT}) + \tilde{\mathcal{O}}(A^{1/3} B^{2/3} K^{1/3} T^{2/3}) + \tilde{\mathcal{O}}(K^{2/3} T^{1/3}/\lambda_{\min}^2) + O(R_{\mathcal{A}}WV_{\text{in}}).$$

721 Union-bounding the windowed self-normalized event, the prefix subspace concentration, and
 722 the noise event at total δ yields (4). For fixed K , the leading term is $\tilde{\mathcal{O}}(r\sqrt{T})$, the probe
 723 term is $\tilde{\mathcal{O}}(T^{2/3})$, and the burn-in term $\tilde{\mathcal{O}}(T^{1/3})$ is lower order. When $V_{\text{in}} = 0$, the third
 724 term vanishes and we recover Corollary 4.3. $\square$

725 B.6 Proof of Corollary 4.4

726 By Theorem B.8, with probability $\geq 1 - \delta$ every eigenvalue of $\widehat{M}_k$ is within $\tau_k^{\text{rank}} :=$
 727 $2R_X\sqrt{\log(2d/\delta)/m_k}$ of the corresponding eigenvalue of $\bar{M}_k^{\text{probe}} + \widehat{B}$ (Weyl's inequality ap-
 728 plied to the matrix in (6)). The population spectrum has exactly r eigenvalues $\geq \lambda_{\min}$ and
 729 the remaining $d - r$ are zero (Prop. B.9). Under the eigengap hypothesis $\lambda_{\min} \geq 4\tau_k^{\text{rank}}$,
 730 thresholding at $2\tau_k^{\text{rank}}$ returns exactly r indices with probability $\geq 1 - \delta$. Condition on this
 731 event; the regret analysis of Theorem 4.1 applies verbatim with the estimated rank equal to
 732 the true r , so (4) holds with an extra δ in the union bound. $\square$

733 B.7 Proof of Proposition 4.5 (SPSC-Adaptive)

734 The detector statistic of Alg. 2 compares two non-overlapping rolling-window lifted-moment
 735 estimators of length W_{det} , $\widehat{M}_t^{\text{recent}}$ and $\widehat{M}_t^{\text{past}}$, via $S_t := \|\widehat{M}_t^{\text{recent}} - \widehat{M}_t^{\text{past}}\|_{\text{op}}$, and triggers
 736 a reset whenever $S_t > 2\eta_{\text{det}}$, where $\eta_{\text{det}} := C_{\text{sub}}\sqrt{\log(dT/\delta_{\text{FA}})/W_{\text{det}}}$ is the calibration
 737 radius. The proof has two parts: bounding the false-alarm probability under H_0 (no change
 738 inside the comparison window) and bounding the detection delay under H_1 (a change of
 739 size $\Delta_k \geq 2b$ has occurred), and then propagating the resulting boundary error through
 740 Theorem 4.1.

741 **(i) False-alarm control under H_0 .** Fix a round t and condition on H_0 : the predictable
 742 second moment is constant across the past and recent windows that produced $\widehat{M}_t^{\text{past}}$ and
 743 $\widehat{M}_t^{\text{recent}}$. Apply Theorem B.8 (matrix Bernstein for the lifted probe estimator) to each
 744 window separately at confidence $\delta_{\text{FA}}/(2T)$ to get

$$\|\widehat{M}_t^{\text{recent}} - \bar{M}_t^{\text{recent}}\|_{\text{op}} \leq \eta_{\text{det}}, \quad \|\widehat{M}_t^{\text{past}} - \bar{M}_t^{\text{past}}\|_{\text{op}} \leq \eta_{\text{det}},$$

each with probability at least $1 - \delta_{\text{FA}}/(2T)$, where $\bar{M}_t^{\text{recent}}$ and $\bar{M}_t^{\text{past}}$ are the predictable averages on the corresponding windows. Under H_0 they coincide, so by the triangle inequality $S_t \leq \|\bar{M}_t^{\text{recent}} - \bar{M}_t^{\text{recent}}\|_{\text{op}} + \|\bar{M}_t^{\text{past}} - \bar{M}_t^{\text{past}}\|_{\text{op}} \leq 2\eta_{\text{det}}$ with probability at least $1 - \delta_{\text{FA}}/T$, i.e. $\Pr(S_t > 2\eta_{\text{det}} \mid H_0) \leq \delta_{\text{FA}}/T$. A union bound over $t \in [T]$ gives total false-alarm probability at most δ_{FA} .

(ii) Detection delay under H_1 . Let τ_k be a true change point with eigengap $\Delta_k := \|P_k^* - P_{k-1}^*\|_{\text{op}} \geq 2b$ for the threshold $b > 2\eta_{\text{det}}$. Take $t = \tau_k + W_{\text{det}}$: the recent window now lies entirely in segment k while the past window lies entirely in segment $k-1$. By the segment-factorization of the predictable probe moment (Proposition B.9) and probe-time excitation (Lemma B.1), $\|\bar{M}_t^{\text{recent}} - \bar{M}_t^{\text{past}}\|_{\text{op}} \geq \lambda_{\min} \Delta_k \geq 2b\lambda_{\min}$. By the same matrix-Bernstein concentration as in (i), $S_t \geq \|\bar{M}_t^{\text{recent}} - \bar{M}_t^{\text{past}}\|_{\text{op}} - 2\eta_{\text{det}} \geq 2b\lambda_{\min} - 2\eta_{\text{det}} > 2\eta_{\text{det}}$ with probability at least $1 - \delta_{\text{FA}}/T$ (using $b > 2\eta_{\text{det}}$ after absorbing $\lambda_{\min}$ into b). Therefore the detector triggers no later than $\hat{\tau}_k = \tau_k + W_{\text{det}}$, giving detection delay $D_k \leq W_{\text{det}}$.

(iii) Boundary-error propagation. On the detector good event $\mathcal{E}_{\text{det}} = \{\text{no false alarm and every change detected within delay } D_k \leq W_{\text{det}}\}$, which holds with probability at least $1 - \delta_{\text{FA}}$ by parts (i)-(ii), the estimated boundaries $\hat{\tau}_k$ satisfy $|\hat{\tau}_k - \tau_k| \leq W_{\text{det}}$. Splitting the horizon at $\hat{\tau}_k$ and applying Theorem 4.1 on each estimated segment, the only difference from the oracle-boundary analysis is the contribution of the rounds in $[\tau_k, \hat{\tau}_k)$ where the algorithm is still using the old basis $\hat{U}_{\tau_k-1}$ on data drawn from segment k . Each such round contributes at most $2R_{\mathcal{A}}S_w$ to the instantaneous regret, so $\sum_k 2R_{\mathcal{A}}S_w D_k \leq 2R_{\mathcal{A}}S_w K W_{\text{det}} = O(KW_{\text{det}})$. Combining,

$$\text{DynReg}_T^{(c)}(\text{Alg. 2}) \leq \text{DynReg}_T^{(c)}(\text{Alg. 1}) + O(KW_{\text{det}})$$

on $\mathcal{E}_{\text{det}}$, which is the claim. The total failure probability is $\delta + \delta_{\text{FA}}$, absorbed into the overall δ budget by re-tuning constants. $\square$

C Probe distributions: scaled sphere and Gaussian truncation

The analysis in §B.2 uses scaled-sphere probes $u_t = \sqrt{d}v_t$, $v_t \sim \text{Unif}(\mathbb{S}^{d-1})$, which satisfy $\|u_t\|_2 = \sqrt{d}$ exactly. No truncation is required: the matrix-Bernstein envelope $R_X = dR_s + S_w^2$ holds deterministically given the noise-truncation event of Lemma B.7.

In our implementation, probes are generated via $u \leftarrow \sqrt{d}\tilde{u}/\|\tilde{u}\|$ for $\tilde{u} \sim \mathcal{N}(0, I_d)$, which is exactly the scaled-sphere distribution.

For isotropic-Gaussian probes (no rescaling), $u_t \sim \mathcal{N}(0, I_d)$, the Laurent–Massart χ^2 tail [Laurent and Massart, 2000, Lem. 1] gives $\Pr(\|u_t\| > L) \leq \delta/T$ at $L = \sqrt{2d \log(2T/\delta)}$; union-bounding over $|\mathcal{T}_{\text{probe}}| \leq T$ gives $\Pr(\bigcap_t \{\|u_t\| \leq L\}) \geq 1 - \delta$. On this event the results of §B.2 apply with $\|u_t\| \leq L$ (replacing $\sqrt{d}$), adding only a $\log(T/\delta)$ factor to R_X and C_{sub} that is absorbed in $\tilde{\mathcal{O}}(\cdot)$.

D Algorithms: pseudocode and adaptive variant

This appendix gives the full pseudocode for SPSC (oracle boundaries, Alg. 1) and SPSC-Adaptive (online change-point detection, Alg. 2), and explains the adaptive design choices (detector form, threshold, recovery phase, burn-in) that tie the tuning back to the guarantee of Proposition 4.5.

Algorithm 1 SPSC: Single-Play Subspace-Calibrated Optimism (oracle boundaries)

```

1: Input: segments  $\{\mathcal{I}_k\}_{k=1}^K$ , probe schedule  $\mathcal{T}_{\text{probe}}$ ,  $\lambda > 0$ , probe distribution  $Q$ , rank  $r$ ,
   window  $W$ .
2: Initialize  $\hat{U}_0 \in \mathbb{R}^{d \times r}$  arbitrarily with orthonormal columns.
3: for  $k = 1, \dots, K$  do ▷ outer loop over segments
4:    $M_{\text{acc}} \leftarrow 0$ ;  $N_k \leftarrow 0$ ;  $\hat{U}_t \leftarrow \hat{U}_{t-1}$ . ▷ reset segment-local accumulator and probe
   counter
5:   for  $t \in \mathcal{I}_k$  do
6:     if  $t \in \mathcal{T}_{\text{probe}}$  then ▷ probe round
7:       draw  $u_t \sim Q$ ; play  $x_t = u_t$ ; observe  $y_t$ .
8:        $s_t \leftarrow y_t^2 - \hat{\sigma}^2$ ;  $G_t \leftarrow \mathcal{K}^{-1}(s_t u_t u_t^\top)$ .
9:        $M_{\text{acc}} \leftarrow M_{\text{acc}} + G_t$ ;  $N_k \leftarrow N_k + 1$ ;  $\hat{U}_t \leftarrow \text{TopEig}_r(M_{\text{acc}}/N_k)$ .
10:    else ▷ exploitation round
11:       $\mathcal{W}_t \leftarrow \{s \in \mathcal{I}_k \setminus \mathcal{T}_{\text{probe}} : t-W \leq s < t\}$ ;  $z_t(x) \leftarrow \hat{U}_{t-1}^\top x$ .
12:       $\tilde{V}_t \leftarrow \lambda I_r + \sum_{s \in \mathcal{W}_t} z_t(x_s) z_t(x_s)^\top$ ,  $\tilde{b}_t \leftarrow \sum_{s \in \mathcal{W}_t} z_t(x_s) y_s$ .
13:       $\hat{a}_t \leftarrow \tilde{V}_t^{-1} \tilde{b}_t$ ; choose  $x_t = \arg \max_{x \in \mathcal{A}_t} \{z_t(x)^\top \hat{a}_t + \beta_t^{(r,W)} \|z_t(x)\|_{\tilde{V}_t^{-1}} +$ 
         $\gamma_t \|x\|_2\}$ .
14:      observe  $y_t$ ;  $\hat{U}_t \leftarrow \hat{U}_{t-1}$ .
15:    end if
16:  end for
17: end for

```

785 D.2 SPSC-Adaptive (unknown boundaries)

786 SPSC-Adaptive recovers segment boundaries from probe data, removing the oracle assumption
 787 of Alg. 1.

788 **Detector.** At each probe round the algorithm maintains two non-overlapping rolling-
 789 window estimators of the lifted second moment: $\hat{M}_t^{\text{recent}}$ over the last n probes and $\hat{M}_t^{\text{past}}$
 790 over the segment-accumulated probes preceding that window. The statistic

$$S_t := \|\hat{M}_t^{\text{recent}} - \hat{M}_t^{\text{past}}\|_{\text{op}} \quad (14)$$

791 is compared to a threshold b ; whenever $S_t > b$ the algorithm declares a change and enters
 792 a RECOVERY phase.

793 **Recovery phase.** Recovery deploys m_{relearn} consecutive probe rounds with no exploitation.
 794 The block-probe design serves two purposes: (i) by Proposition B.9, m_{relearn}
 795 back-to-back probes produce a subspace estimate $\hat{U}$ with projector error $\|\hat{P} - P^\star\|_{\text{op}} =$
 796 $O(1/\sqrt{m_{\text{relearn}}})$ on a $1 - \delta$ event; (ii) by zeroing out stale buffers it prevents the windowed
 797 ridge estimator from carrying information from the previous regime into the new one. After
 798 m_{relearn} rounds the algorithm transitions back to NORMAL probe-exploitation interleaving
 799 with probe rate μ .

800 **Threshold calibration and detection delay.** Under H_0 (no change), matrix Freedman
 801 applied to the two windows gives $\Pr(S_t > 2\eta_{\text{det}}) \leq \delta_{\text{FA}}/T$ with $\eta_{\text{det}} = C_{\text{sub}} \sqrt{\log(dT/\delta_{\text{FA}})/n}$;
 802 choosing $b = 2\eta_{\text{det}}$ and union-bounding gives at most δ_{FA} false alarms across the horizon.
 803 Under H_1 at time τ_k , a separation $\Delta_k \geq 2b$ forces a fire within delay $D_{\text{max}} = 2n/\mu + \tau_{\text{burn}}$,
 804 where the $2n/\mu$ term is the time for the recent window to refill with post-change probes, and
 805 τ_{burn} is a short burn-in period after recovery that prevents the detector from firing on its
 806 own freshly-reset state. The overall regret overhead is $2R_{\mathcal{A}} S_w \sum_k D_k = O(KW_{\text{det}})$, which
 807 is lower-order in the rate of Proposition 4.5.

Algorithm 2 SPSC-Adaptive: unknown boundaries via CUSUM-style detection

```
1: Input: probe rate  $\mu \in (0, 1)$ , detection window  $n$ , threshold  $b$ , burn-in  $\tau_{\text{burn}}$ , relearning
   budget  $m_{\text{relearn}}$ ,  $W$ ,  $\lambda$ .
2: phase  $\leftarrow$  RECOVERY; rec  $\leftarrow$  0;  $\widehat{M}_{\text{acc}} \leftarrow 0$ .
3: for  $t = 1, \dots, T$  do
4:   if phase = RECOVERY then  $\triangleright$  block-probe to rebuild subspace
5:     probe: draw  $u_t$ , play  $x_t = u_t$ , observe  $y_t$ ; update  $\widehat{M}_{\text{acc}}$ ; rec  $+= 1$ .
6:     if rec =  $m_{\text{relearn}}$  then
7:        $\widehat{U} \leftarrow \text{TopEig}_r(\widehat{M}_{\text{acc}}/m_{\text{relearn}})$ ; reset buffers; phase  $\leftarrow$  NORMAL.
8:     end if
9:   else  $\triangleright$  interleave probes at rate  $\mu$ 
10:    if Bernoulli( $\mu$ ) probe round then
11:      update lifted estimator  $\widehat{M}_{\text{acc}}$  and refresh rolling windows.
12:      if  $S_t > b$  (Eq. 14) and detector is armed then
13:        phase  $\leftarrow$  RECOVERY; rec  $\leftarrow$  0; reset.
14:      end if
15:    else
16:      windowed projected ridge-UCB step as in Alg. 1 (exploit branch).
17:    end if
18:  end if
19: end for
```

808 **Tuning used in experiments.** Across all experiments reported in §5 and the appendix
809 we use $\mu = 0.1$, $n = W_{\text{det}} = 50$, $\tau_{\text{burn}} = 100$, $m_{\text{relearn}} = 30$, and b set from the CUSUM
810 threshold `cusum_threshold` = 3.0. These are held fixed across datasets and (d, r) cells; no
811 per-dataset retuning is done. Sensitivity to the probe rate μ and the detection window n is
812 reported in §G.

813 E Experimental setup

814 **Synthetic phase-transition benchmark.** $\theta_t = B_k^* w_t$ with piecewise-constant orthonormal
815 $B_k^* \in \mathbb{R}^{d \times r}$. Per segment, latent state evolves as $w_t = A_k w_{t-1} + \eta_{t-1}$ with spectral radius
816 $\rho(A_k) = 0.99$ and Gaussian innovation $\eta_{t-1} \sim \mathcal{N}(0, 0.04^2 I_r)$. Reward noise $\varepsilon_t \sim \mathcal{N}(0, 0.09)$;
817 40 actions per round drawn i.i.d. uniformly on the unit sphere. $K=10$ segments of equal
818 length over $T=5,000$. Probe distribution Q isotropic Gaussian, probe cost $c=0.1$, probe
819 period 50.

820 **UCI datasets (Coverttype, Pendigits, Satimage).** Features are the raw attribute
821 vectors $\phi(x) \in \mathbb{R}^{d_0}$ reduced to ambient dimension d via Gaussian random projection $R \in$
822 $\mathbb{R}^{d \times d_0}$. For Coverttype, segments correspond to disjoint size- K cover-type subsets of the 7
823 classes; for Pendigits and Satimage, to disjoint digit/class subsets, $K=10$. Actions at round
824 t are $|\mathcal{A}_t| = 40$ examples drawn from the active segment’s class pool. Reward is centered
825 target label. $\lambda=0.01$, probe period 10, window $W=400$, $\delta=0.05$, $c=0.02$.

826 **Pixelized MNIST / Fashion-MNIST.** Images flattened, pixel-normalized, then pro-
827 jected to d via random projection. Segments are class-subset swaps as above. Same hyper-
828 parameters as UCI datasets.

829 **MovieLens-100K with real ratings.** Standard MovieLens-100K user-item matrix cen-
830 tered by per-item mean. Features $\phi(\text{user}, \text{item})$ are item-side latent factors from a rank- d
831 SVD of the observed-entry mask; reward is the raw user rating (1–5), centered. Segments
832 correspond to disjoint time-bucketed user cohorts, $K=10$. This is a *fully real* benchmark
833 (we do *not* generate rewards from a low-rank surrogate), so the data is only approximately
834 low-rank and provides a stringent stress test. $T=5,000$, same hyperparameters.

Warfarin clinical dosing. IWPC cohort statistics ($\sim 5,700$ patients); $d=93$ features (demographics, VKORC1/CYP2C9 genotype, medications, interactions). Reward is negative absolute deviation from the therapeutic dose, $K=8$ patient-cohort segments, $T=5,000$, $\lambda=1.0$, probe period 10, $W=200$, $c=0.1$, 10 seeds.

Baseline hyperparameters. All baselines inherit λ, δ from SPSC when applicable. D-LinUCB uses discount factor 0.998; SW-LinUCB uses window W ; Restart-LinUCB restarts every T/K rounds (i.e. oracle boundaries); LinTS/SW-LinTS use posterior variance $\sigma^2 = \delta^2$. Oracle-LinUCB knows the true B_k^* and runs LinUCB in the projected $\mathbb{R}^r$.

Subspace-aware baselines (LowOFUL, VOFUL): favorable adaptation. Both LowOFUL and VOFUL were designed for a *fixed* low-rank representation, so running them in our piecewise setting requires a choice for the subspace they consume. To avoid conflating subspace acquisition with exploitation, we give these baselines the same refreshed projector $\hat{U}$ that SPSC uses (recomputed every 20 rounds after a 30-round warm-up), so every subspace-aware method sees an identical low-rank representation at every round. This is a *favorable* adaptation, since it removes representation quality as a possible source of SPSC’s empirical gains, and any remaining gap is attributable to SPSC’s windowed head and drift handling rather than to better subspace estimation. Running LowOFUL/VOFUL with their own stationary subspace estimators (not shown) yields strictly worse baseline numbers and larger SPSC margins.

Implementation and compute. All experiments use NumPy/SciPy only. A full (d, r) grid (9 cells, 10 seeds, nine per-cell baselines plus Oracle-LinUCB and the two SPSC variants; BOSS and Jedra appear only in the §5.6 grid) takes 6–20 hours on a single 16-core CPU. Covertypes is the longest ($T=10,000$).

F Per-cell tables: every dataset, every cell

Tables 8–15 give the full per-cell results (mean $\pm$ SE costed regret over 10 seeds) for every method on every (d, r) cell. Cells where at least one SPSC variant beats every non-oracle baseline are highlighted **green**; the best non-oracle baseline per cell is underlined.

Table 8: **Covertypes** per-cell results. Mean $\pm$ SE costed regret over 10 seeds. Green = SPSC variant beats every non-oracle baseline; underline = best non-oracle. $T=10,000$, $K=4$. *Reading note (applies to all per-cell tables in this appendix):* rank-insensitive methods (LinUCB, SW-LinUCB, D-LinUCB, Restart-LinUCB, LinTS, SW-LinTS) act in ambient space; their regret depends on d but not on r , hence bit-identical values across the rows of each d -block.

d	r	Oracle	SPSC	SPSC-Adp	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
5	1	2013 $\pm$ 138	2655 $\pm$ 85	2114 $\pm$ 44	3748 $\pm$ 162	3736 $\pm$ 164	2379 $\pm$ 136	2651 $\pm$ 110	2502 $\pm$ 98	2661 $\pm$ 124	2694 $\pm$ 126	3633 $\pm$ 159	3064 $\pm$ 113
	55	1	718 $\pm$ 51	1296 $\pm$ 21	1262 $\pm$ 32	1429 $\pm$ 32	1436 $\pm$ 32	1366 $\pm$ 44	1331 $\pm$ 20	1346 $\pm$ 21	1329 $\pm$ 21	1336 $\pm$ 21	1462 $\pm$ 25
10	1	2768 $\pm$ 29	4307 $\pm$ 62	4161 $\pm$ 63	5067 $\pm$ 83	5167 $\pm$ 130	5122 $\pm$ 61	4403 $\pm$ 41	4599 $\pm$ 57	4348 $\pm$ 37	4465 $\pm$ 49	5591 $\pm$ 78	5486 $\pm$ 73
	20	4004 $\pm$ 55	5769 $\pm$ 54	5786 $\pm$ 84	7120 $\pm$ 93	7001 $\pm$ 72	7538 $\pm$ 81	5703 $\pm$ 34	5971 $\pm$ 40	5669 $\pm$ 40	5800 $\pm$ 50	7825 $\pm$ 41	7581 $\pm$ 44
30	1	5202 $\pm$ 48	6961 $\pm$ 95	7051 $\pm$ 81	8890 $\pm$ 110	8770 $\pm$ 93	9527 $\pm$ 96	6653 $\pm$ 34	6928 $\pm$ 30	6594 $\pm$ 51	6743 $\pm$ 34	9615 $\pm$ 105	9345 $\pm$ 90
	105	1	554 $\pm$ 30	1023 $\pm$ 29	1010 $\pm$ 21	1134 $\pm$ 41	1124 $\pm$ 44	1026 $\pm$ 28	1075 $\pm$ 32	1082 $\pm$ 33	1073 $\pm$ 31	1075 $\pm$ 33	1109 $\pm$ 35
10	1	2154 $\pm$ 27	3452 $\pm$ 45	3405 $\pm$ 40	3775 $\pm$ 34	3758 $\pm$ 48	3869 $\pm$ 36	3920 $\pm$ 36	3959 $\pm$ 36	3909 $\pm$ 37	3943 $\pm$ 35	4198 $\pm$ 40	4168 $\pm$ 37
	20	3233 $\pm$ 36	4600 $\pm$ 62	4507 $\pm$ 46	5297 $\pm$ 67	5325 $\pm$ 68	5621 $\pm$ 43	5302 $\pm$ 40	5383 $\pm$ 32	5313 $\pm$ 30	5355 $\pm$ 35	5826 $\pm$ 42	5734 $\pm$ 42
30	1	4165 $\pm$ 23	5529 $\pm$ 71	5450 $\pm$ 67	6476 $\pm$ 33	6556 $\pm$ 51	6989 $\pm$ 54	6360 $\pm$ 38	6473 $\pm$ 33	6324 $\pm$ 42	6400 $\pm$ 37	7085 $\pm$ 48	7029 $\pm$ 46

Table 9: **Pendigits** per-cell results. $T=5,000$, $K=10$, 10 seeds.

d	r	Oracle	SPSC	SPSC-Adp	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
5	1	39.0 $\pm$ 2.0	2914 $\pm$ 159	5480 $\pm$ 391	9536 $\pm$ 199	9623 $\pm$ 138	6606 $\pm$ 77	1090 $\pm$ 9	1267 $\pm$ 8	1022 $\pm$ 14	1073 $\pm$ 15	6629 $\pm$ 42	4630 $\pm$ 41
	55	1	18.0 $\pm$ 1.0	4019 $\pm$ 79	4073 $\pm$ 200	5913 $\pm$ 46	5939 $\pm$ 64	4542 $\pm$ 48	3863 $\pm$ 11	4056 $\pm$ 9	3620 $\pm$ 7	3658 $\pm$ 9	4343 $\pm$ 16
10	1	951 $\pm$ 5	2774 $\pm$ 58	2682 $\pm$ 93	3703 $\pm$ 71	3655 $\pm$ 67	4246 $\pm$ 28	3863 $\pm$ 11	4056 $\pm$ 9	3620 $\pm$ 7	3658 $\pm$ 9	4343 $\pm$ 16	4648 $\pm$ 9
	20	1816 $\pm$ 4	2803 $\pm$ 60	2490 $\pm$ 56	3689 $\pm$ 54	3854 $\pm$ 58	4271 $\pm$ 39	3863 $\pm$ 11	4056 $\pm$ 9	3620 $\pm$ 7	3658 $\pm$ 9	4343 $\pm$ 16	4648 $\pm$ 9
30	1	2534 $\pm$ 6	3051 $\pm$ 62	2776 $\pm$ 45	3827 $\pm$ 45	3818 $\pm$ 39	4329 $\pm$ 20	3863 $\pm$ 11	4056 $\pm$ 9	3620 $\pm$ 7	3658 $\pm$ 9	4343 $\pm$ 16	4648 $\pm$ 9
	105	1	11.0 $\pm$ 0.0	3011 $\pm$ 88	3120 $\pm$ 134	4254 $\pm$ 114	4277 $\pm$ 107	3051 $\pm$ 89	3297 $\pm$ 6	3400 $\pm$ 6	3152 $\pm$ 6	3184 $\pm$ 5	3117 $\pm$ 10
10	1	666 $\pm$ 3	2066 $\pm$ 64	1693 $\pm$ 38	2927 $\pm$ 74	2816 $\pm$ 55	3027 $\pm$ 43	3297 $\pm$ 6	3400 $\pm$ 6	3152 $\pm$ 6	3184 $\pm$ 5	3117 $\pm$ 10	3557 $\pm$ 4
	20	1242 $\pm$ 3	2067 $\pm$ 62	1571 $\pm$ 67	2840 $\pm$ 41	2835 $\pm$ 35	3123 $\pm$ 18	3297 $\pm$ 6	3400 $\pm$ 6	3152 $\pm$ 6	3184 $\pm$ 5	3117 $\pm$ 10	3557 $\pm$ 4
30	1	1745 $\pm$ 4	2247 $\pm$ 58	1722 $\pm$ 58	2826 $\pm$ 75	2839 $\pm$ 56	3283 $\pm$ 19	3297 $\pm$ 6	3400 $\pm$ 6	3152 $\pm$ 6	3184 $\pm$ 5	3117 $\pm$ 10	3557 $\pm$ 4

Table 10: **Satimage** per-cell results. $T=5,000$, $K=10$, 10 seeds.

d	r	Oracle	SPSC	SPSC-Adp	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
5	1	35.0 $\pm$ 2.0	1480 $\pm$ 88	1513 $\pm$ 123	2739 $\pm$ 17	2704 $\pm$ 18	2660 $\pm$ 24	655 $\pm$ 7	778 $\pm$ 6	629 $\pm$ 6	680 $\pm$ 6	2397 $\pm$ 14	1675 $\pm$ 16
55	1	18.0 $\pm$ 1.0	2439 $\pm$ 81	2015 $\pm$ 75	2913 $\pm$ 11	2911 $\pm$ 8	3450 $\pm$ 52	3051 $\pm$ 6	3201 $\pm$ 9	2779 $\pm$ 3	2848 $\pm$ 9	3403 $\pm$ 11	3781 $\pm$ 10
	10	597 $\pm$ 3	1343 $\pm$ 41	1073 $\pm$ 49	2294 $\pm$ 99	2288 $\pm$ 96	2689 $\pm$ 62	3051 $\pm$ 6	3201 $\pm$ 9	2779 $\pm$ 3	2848 $\pm$ 9	3403 $\pm$ 11	3781 $\pm$ 10
	20	1218 $\pm$ 4	1506 $\pm$ 34	1201 $\pm$ 33	2455 $\pm$ 106	2479 $\pm$ 65	2879 $\pm$ 55	3051 $\pm$ 6	3201 $\pm$ 9	2779 $\pm$ 3	2848 $\pm$ 9	3403 $\pm$ 11	3781 $\pm$ 10
	30	1725 $\pm$ 8	1900 $\pm$ 38	1635 $\pm$ 40	2641 $\pm$ 54	2772 $\pm$ 47	3085 $\pm$ 53	3051 $\pm$ 6	3201 $\pm$ 9	2779 $\pm$ 3	2848 $\pm$ 9	3403 $\pm$ 11	3781 $\pm$ 10
105	1	16.0 $\pm$ 1.0	2484 $\pm$ 59	2333 $\pm$ 108	3638 $\pm$ 7	3655 $\pm$ 12	3911 $\pm$ 47	3572 $\pm$ 8	3671 $\pm$ 9	3441 $\pm$ 6	3484 $\pm$ 10	3600 $\pm$ 9	3891 $\pm$ 7
	10	538 $\pm$ 3	1377 $\pm$ 44	1112 $\pm$ 49	2431 $\pm$ 87	2361 $\pm$ 78	2703 $\pm$ 40	3572 $\pm$ 8	3671 $\pm$ 9	3441 $\pm$ 6	3484 $\pm$ 10	3600 $\pm$ 9	3891 $\pm$ 7
	20	1098 $\pm$ 5	1454 $\pm$ 33	1182 $\pm$ 34	2229 $\pm$ 57	2346 $\pm$ 46	2697 $\pm$ 39	3572 $\pm$ 8	3671 $\pm$ 9	3441 $\pm$ 6	3484 $\pm$ 10	3600 $\pm$ 9	3891 $\pm$ 7
	30	1569 $\pm$ 5	1769 $\pm$ 32	1506 $\pm$ 33	2525 $\pm$ 42	2587 $\pm$ 80	2722 $\pm$ 25	3572 $\pm$ 8	3671 $\pm$ 9	3441 $\pm$ 6	3484 $\pm$ 10	3600 $\pm$ 9	3891 $\pm$ 7

Table 11: **Fashion-MNIST** per-cell results. $T=5,000$, $K=10$, 10 seeds.

d	r	Oracle	SPSC	SPSC-Adp	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
55	5	539 $\pm$ 6	3773 $\pm$ 91	3792 $\pm$ 58	4487 $\pm$ 103	4475 $\pm$ 118	4719 $\pm$ 93	4169 $\pm$ 9	4324 $\pm$ 11	3801 $\pm$ 6	3868 $\pm$ 12	5183 $\pm$ 11	5357 $\pm$ 15
	10	945 $\pm$ 5	3406 $\pm$ 73	3377 $\pm$ 92	4425 $\pm$ 126	4471 $\pm$ 142	4740 $\pm$ 59	4169 $\pm$ 9	4324 $\pm$ 11	3801 $\pm$ 6	3868 $\pm$ 12	5183 $\pm$ 11	5357 $\pm$ 15
	20	1676 $\pm$ 6	3450 $\pm$ 62	3162 $\pm$ 81	4677 $\pm$ 119	4708 $\pm$ 75	4924 $\pm$ 56	4169 $\pm$ 9	4324 $\pm$ 11	3801 $\pm$ 6	3868 $\pm$ 12	5183 $\pm$ 11	5357 $\pm$ 15
105	5	399 $\pm$ 3	3200 $\pm$ 88	3208 $\pm$ 60	3365 $\pm$ 71	3522 $\pm$ 92	3688 $\pm$ 81	4152 $\pm$ 8	4236 $\pm$ 7	4016 $\pm$ 8	4034 $\pm$ 10	4376 $\pm$ 9	4496 $\pm$ 11
	10	757 $\pm$ 6	2927 $\pm$ 76	2623 $\pm$ 84	3331 $\pm$ 42	3478 $\pm$ 51	3645 $\pm$ 65	4152 $\pm$ 8	4236 $\pm$ 7	4016 $\pm$ 8	4034 $\pm$ 10	4376 $\pm$ 9	4496 $\pm$ 11
	20	1439 $\pm$ 4	2855 $\pm$ 75	2528 $\pm$ 59	3559 $\pm$ 80	3564 $\pm$ 45	3797 $\pm$ 38	4152 $\pm$ 8	4236 $\pm$ 7	4016 $\pm$ 8	4034 $\pm$ 10	4376 $\pm$ 9	4496 $\pm$ 11
200	5	393 $\pm$ 3	2949 $\pm$ 52	2877 $\pm$ 89	3134 $\pm$ 72	2869 $\pm$ 58	2948 $\pm$ 47	4099 $\pm$ 7	4158 $\pm$ 8	4059 $\pm$ 5	4059 $\pm$ 11	4101 $\pm$ 11	4206 $\pm$ 8
	10	780 $\pm$ 4	2703 $\pm$ 52	2732 $\pm$ 101	2929 $\pm$ 43	2731 $\pm$ 63	2967 $\pm$ 26	4099 $\pm$ 7	4158 $\pm$ 8	4059 $\pm$ 5	4059 $\pm$ 11	4101 $\pm$ 11	4206 $\pm$ 8
	20	1290 $\pm$ 5	2654 $\pm$ 52	2582 $\pm$ 67	3215 $\pm$ 56	3196 $\pm$ 61	3341 $\pm$ 44	4099 $\pm$ 7	4158 $\pm$ 8	4059 $\pm$ 5	4059 $\pm$ 11	4101 $\pm$ 11	4206 $\pm$ 8

Table 12: **MNIST** per-cell results. $T=5,000$, $K=10$, 10 seeds.

d	r	Oracle	SPSC	SPSC-Adp	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
55	5	666 $\pm$ 5	5076 $\pm$ 88	5223 $\pm$ 109	6437 $\pm$ 152	6646 $\pm$ 62	6650 $\pm$ 116	5889 $\pm$ 16	6154 $\pm$ 12	5426 $\pm$ 10	5533 $\pm$ 9	6958 $\pm$ 13	7362 $\pm$ 20
	10	1149 $\pm$ 9	4648 $\pm$ 77	4968 $\pm$ 93	6392 $\pm$ 114	6191 $\pm$ 118	6623 $\pm$ 77	5889 $\pm$ 16	6154 $\pm$ 12	5426 $\pm$ 10	5533 $\pm$ 9	6958 $\pm$ 13	7362 $\pm$ 20
	20	2196 $\pm$ 11	4745 $\pm$ 76	4549 $\pm$ 82	6420 $\pm$ 63	6352 $\pm$ 120	6745 $\pm$ 58	5889 $\pm$ 16	6154 $\pm$ 12	5426 $\pm$ 10	5533 $\pm$ 9	6958 $\pm$ 13	7362 $\pm$ 20
105	5	582 $\pm$ 5	4658 $\pm$ 104	4844 $\pm$ 68	5184 $\pm$ 74	5305 $\pm$ 56	5807 $\pm$ 93	5992 $\pm$ 11	6073 $\pm$ 8	5749 $\pm$ 11	5811 $\pm$ 10	5892 $\pm$ 18	6222 $\pm$ 11
	10	1029 $\pm$ 5	4298 $\pm$ 96	4229 $\pm$ 157	4867 $\pm$ 96	4917 $\pm$ 46	5555 $\pm$ 58	5992 $\pm$ 11	6073 $\pm$ 8	5749 $\pm$ 11	5811 $\pm$ 10	5892 $\pm$ 18	6222 $\pm$ 11
	20	1869 $\pm$ 11	4273 $\pm$ 101	4076 $\pm$ 114	5020 $\pm$ 46	4918 $\pm$ 78	5651 $\pm$ 46	5992 $\pm$ 11	6073 $\pm$ 8	5749 $\pm$ 11	5811 $\pm$ 10	5892 $\pm$ 18	6222 $\pm$ 11
200	5	539 $\pm$ 2	4377 $\pm$ 84	4260 $\pm$ 81	4672 $\pm$ 62	4718 $\pm$ 113	4797 $\pm$ 52	5751 $\pm$ 10	5786 $\pm$ 6	5659 $\pm$ 11	5661 $\pm$ 8	5568 $\pm$ 10	5774 $\pm$ 7
	10	907 $\pm$ 5	4114 $\pm$ 66	4023 $\pm$ 113	4184 $\pm$ 63	4375 $\pm$ 75	4818 $\pm$ 42	5751 $\pm$ 10	5786 $\pm$ 6	5659 $\pm$ 11	5661 $\pm$ 8	5568 $\pm$ 10	5774 $\pm$ 7
	20	1675 $\pm$ 6	4052 $\pm$ 59	3828 $\pm$ 69	4379 $\pm$ 34	4413 $\pm$ 50	5022 $\pm$ 27	5751 $\pm$ 10	5786 $\pm$ 6	5659 $\pm$ 11	5661 $\pm$ 8	5568 $\pm$ 10	5774 $\pm$ 7

Table 13: **MovieLens** (fully real ratings) per-cell results. $T=5,000$, $K=10$, 10 seeds.

d	r	Oracle	SPSC	SPSC-Adp	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
55	5	4013 $\pm$ 33	6213 $\pm$ 63	6060 $\pm$ 77	7676 $\pm$ 164	7504 $\pm$ 143	8103 $\pm$ 60	6313 $\pm$ 6	6363 $\pm$ 6	6239 $\pm$ 10	6258 $\pm$ 8	6396 $\pm$ 11	6587 $\pm$ 11
	10	4802 $\pm$ 24	6250 $\pm$ 46	6015 $\pm$ 37	7689 $\pm$ 77	7461 $\pm$ 113	7855 $\pm$ 47	6313 $\pm$ 6	6363 $\pm$ 6	6239 $\pm$ 10	6258 $\pm$ 8	6396 $\pm$ 11	6587 $\pm$ 11
	20	5402 $\pm$ 21	6266 $\pm$ 42	6216 $\pm$ 44	7407 $\pm$ 56	7329 $\pm$ 108	7391 $\pm$ 30	6313 $\pm$ 6	6363 $\pm$ 6	6239 $\pm$ 10	6258 $\pm$ 8	6396 $\pm$ 11	6587 $\pm$ 11
105	5	3970 $\pm$ 33	6535 $\pm$ 56	6350 $\pm$ 73	8276 $\pm$ 126	8292 $\pm$ 89	8340 $\pm$ 35	6809 $\pm$ 5	6850 $\pm$ 4	6783 $\pm$ 4	6793 $\pm$ 6	6743 $\pm$ 13	6874 $\pm$ 12
	10	4951 $\pm$ 37	6560 $\pm$ 43	6297 $\pm$ 55	7801 $\pm$ 173	7927 $\pm$ 154	8200 $\pm$ 61	6809 $\pm$ 5	6850 $\pm$ 4	6783 $\pm$ 4	6793 $\pm$ 6	6743 $\pm$ 13	6874 $\pm$ 12
	20	5697 $\pm$ 23	6636 $\pm$ 32	6566 $\pm$ 40	7562 $\pm$ 103	7688 $\pm$ 90	7674 $\pm$ 27	6809 $\pm$ 5	6850 $\pm$ 4	6783 $\pm$ 4	6793 $\pm$ 6	6743 $\pm$ 13	6874 $\pm$ 12
200	5	3939 $\pm$ 27	6787 $\pm$ 52	6516 $\pm$ 101	8365 $\pm$ 102	8450 $\pm$ 118	8420 $\pm$ 46	7396 $\pm$ 6	7421 $\pm$ 6	7404 $\pm$ 4	7417 $\pm$ 7	7134 $\pm$ 12	7203 $\pm$ 7
	10	4998 $\pm$ 48	6844 $\pm$ 57	6576 $\pm$ 36	8383 $\pm$ 131	8163 $\pm$ 135	8513 $\pm$ 50	7396 $\pm$ 6	7421 $\pm$ 6	7404 $\pm$ 4	7417 $\pm$ 7	7134 $\pm$ 12	7203 $\pm$ 7
	20	5871 $\pm$ 25	6995 $\pm$ 47	6844 $\pm$ 53	7296 $\pm$ 118	7605 $\pm$ 117	7627 $\pm$ 58	7396 $\pm$ 6	7421 $\pm$ 6	7404 $\pm$ 4	7417 $\pm$ 7	7134 $\pm$ 12	7203 $\pm$ 7

Table 14: **Warfarin** ($d=93$, $K=8$, $T=5,000$) per-rank results across $r \in \{1, 2, 3, 5, 10\}$. 10 seeds.

d	r	Oracle	SPSC	SPSC-Adp	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB	LinTS	SW-LinTS
93	1	6.4 $\pm$ 0.7	1482 $\pm$ 49	626 $\pm$ 44	1491 $\pm$ 26	1497 $\pm$ 23	1533 $\pm$ 20	2096 $\pm$ 18	2175 $\pm$ 22	1977 $\pm$ 15	2006 $\pm$ 17	1949 $\pm$ 20	2060 $\pm$ 22
	2	93.9 $\pm$ 1.3	1404 $\pm$ 35	706 $\pm$ 37	1076 $\pm$ 130	918 $\pm$ 127	1140 $\pm$ 25	2096 $\pm$ 18	2175 $\pm$ 22	1977 $\pm$ 15	2006 $\pm$ 17	1949 $\pm$ 20	2060 $\pm$ 22
	3	177 $\pm$ 4	1372 $\pm$ 33	657 $\pm$ 39	683 $\pm$ 97	779 $\pm$ 93	1056 $\pm$ 40	2096 $\pm$ 18	2175 $\pm$ 22	1977 $\pm$ 15	2006 $\pm$ 17	1949 $\pm$ 20	2060 $\pm$ 22
	5	272 $\pm$ 7	1369 $\pm$ 30	769 $\pm$ 37	905 $\pm$ 110	930 $\pm$ 76	1178 $\pm$ 32	2096 $\pm$ 18	2175 $\pm$ 22	1977 $\pm$ 15	2006 $\pm$ 17	1949 $\pm$ 20	2060 $\pm$ 22
	10	569 $\pm$ 4	1481 $\pm$ 19	1014 $\pm$ 35	1441 $\pm$ 63	1680 $\pm$ 59	2078 $\pm$ 19	2096 $\pm$ 18	2175 $\pm$ 22	1977 $\pm$ 15	2006 $\pm$ 17	1949 $\pm$ 20	2060 $\pm$ 22

Table 15: **Open Bandit (ZOZOTOWN)** per-cell results. $T=5,000$, $K=10$, 10 seeds. Mean $\pm$ SE costed regret. Green = SPSC variant beats every non-oracle baseline; underline = best non-oracle. LinTS / SW-LinTS not run on this benchmark.

d	r	Oracle	SPSC	SPSC-Adp	LowOFUL	VOFUL	LR-Rew	SW-Lin	D-Lin	Rst-Lin	LinUCB
55	5	34.2 $\pm$ 0.5	83.0 $\pm$ 2.3	91.1 $\pm$ 4.0	126.2 $\pm$ 1.8	126.7 $\pm$ 1.5	126.8 $\pm$ 1.2	103.0 $\pm$ 0.5	106.6 $\pm$ 0.5	101.9 $\pm$ 0.6	102.2 $\pm$ 0.5
	10	65.2 $\pm$ 0.8	93.9 $\pm$ 1.5	94.1 $\pm$ 1.7	129.3 $\pm$ 2.6	130.2 $\pm$ 1.9	130.9 $\pm$ 1.3	103.0 $\pm$ 0.5	106.6 $\pm$ 0.5	101.9 $\pm$ 0.6	102.2 $\pm$ 0.5
105	5	29.7 $\pm$ 0.5	106.6 $\pm$ 4.8	103.3 $\pm$ 2.9	152.6 $\pm$ 2.3	150.1 $\pm$ 1.9	150.9 $\pm$ 1.6	107.4 $\pm$ 1.1	108.4 $\pm$ 0.4	106.4 $\pm$ 0.5	107.7 $\pm$ 1.0
	10	59.0 $\pm$ 0.7	106.9 $\pm$ 3.0	106.3 $\pm$ 3.2	153.3 $\pm$ 1.0	155.4 $\pm$ 1.8	156.1 $\pm$ 1.5	107.4 $\pm$ 1.1	108.4 $\pm$ 0.4	106.4 $\pm$ 0.5	107.7 $\pm$ 1.0

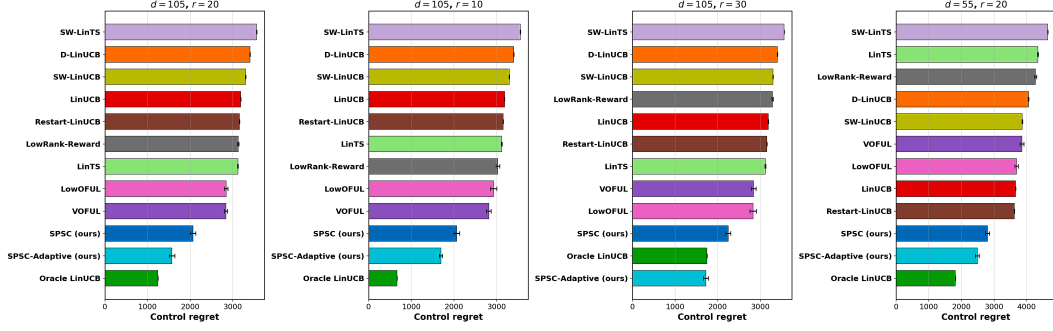


Figure 2: **Pendigits operating-regime grid.** SPSC and SPSC-Adaptive dominate every non-oracle baseline once $d \geq 55$ and $r \geq 10$.

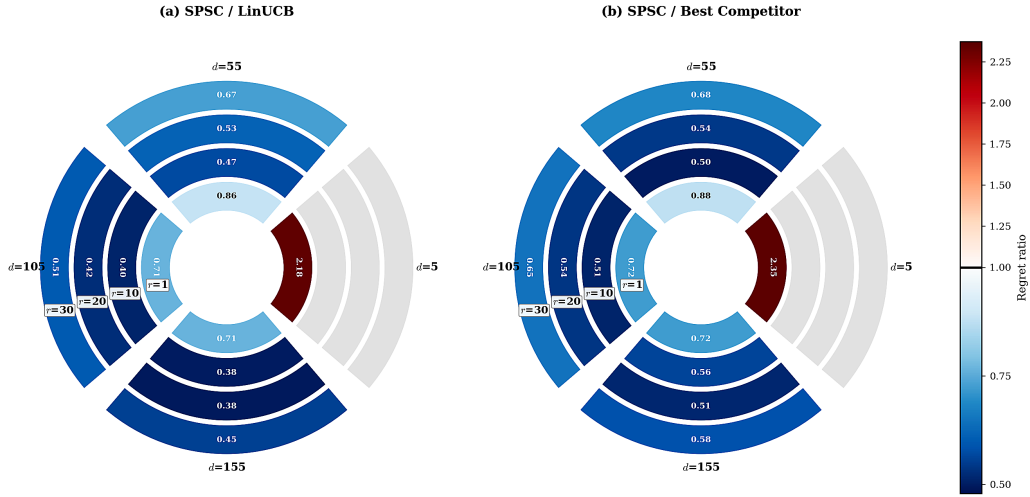


Figure 3: **Satimage operating-regime sunburst.** Regret ratios across the (d, r) grid. Blue cells (ratio < 1) indicate SPSC wins; red cells indicate losses. (a) SPSC vs. LinUCB; (b) SPSC vs. best non-oracle competitor.

Reading the tables. Three qualitative patterns are consistent across datasets. (i) *Rank-insensitive competitors.* LinUCB, SW-LinUCB, D-LinUCB and Restart-LinUCB operate in ambient space and therefore produce the *same* regret in every row of a d block. When a row shows all four of these at identical values, this is by design: it reflects the r -insensitivity of ambient-rate baselines. (ii) *Subspace-aware competitors* (LowOFUL, VOFUL, LowRank-Reward) do improve with d increasing but they assume a fixed subspace and therefore still trail SPSC on segments where the subspace changes. (iii) *Within the synthetic and UCI grids, SPSC-Adaptive typically trails SPSC-Alg. 1 by a small margin;* this gap reflects the detector’s false-alarm budget. The ordering reverses on the clinical benchmarks (most strikingly Vancomycin at $r=1$, where SPSC-Adaptive cuts regret by 37% vs. SPSC-Alg. 1; see Table 19), where the oracle schedule of Alg. 1 is misaligned with the true cohort change points and the detector’s adaptivity wins.

Synthetic (d, r) grid. Table 16 gives the full synthetic grid underlying Figure 1. SPSC wins on 37/40 cells; losses are confined to $d \leq 10$ or $r \leq 1$ (probe-cost dominant).

Table 16: **Synthetic grid (full 40 cells).** Mean costed regret across 10 seeds for 10 methods over $(d, r) \in \{5, 10, 20, 30, 45, 60, 80, 100\} \times \{1, 3, 5, 10, 15, 20\}$ with $r < d$, $T=5,000$, $K=10$, probe period 50, window $W=400$, $\lambda=0.01$. **Methods:** SPSC (Algorithm 1, ours); **Adpt** = SPSC-Adaptive (Algorithm 2, ours); LinUCB; **Oracle** = Oracle-LinUCB (given the true subspace); **SW-Lin** = SW-LinUCB; **D-Lin** = D-LinUCB; **Rst** = Restart-LinUCB; **LR-Rew** = LowRank-Reward; LowOFUL; VOFUL. S/L = SPSC/LinUCB; S/Best = SPSC/ $\min\{\text{non-oracle, non-Adpt baselines}\}$. **Verdict.** “SPSC” if $S/L < 1$ (SPSC has lower regret than LinUCB); “LinUCB” otherwise. Bold = SPSC verdict. Counts: **37 SPSC**, **3 LinUCB**; the 3 LinUCB cells are all in the small- d corner $d \leq 10$ with r close to d , where the probe cost dominates the dimensionality savings.

d	r	SPSC	Adpt	LinUCB	Oracle	SW-Lin	D-Lin	Rst	LR-Rew	LowOFUL	VOFUL	S/L	S/Best	verdict
5	1	355	344	203	8	213	234	206	416	514	553	1.742	1.742	LinUCB
5	3	339	384	221	134	221	254	217	629	930	889	1.529	1.563	LinUCB
5	10	270	289	255	24	260	271	258	254	317	315	1.055	1.060	LinUCB
10	3	389	422	399	130	391	428	386	456	629	595	0.976	1.008	SPSC
10	5	465	525	487	254	482	523	471	660	800	823	0.955	0.988	SPSC
20	1	231	228	251	21	251	253	248	179	238	240	0.923	1.290	SPSC
20	3	353	377	456	134	449	467	451	344	449	475	0.774	1.026	SPSC
20	5	420	453	565	232	555	581	558	468	575	578	0.744	0.898	SPSC
20	10	631	699	800	498	780	840	784	804	946	975	0.789	0.809	SPSC
20	15	805	901	899	736	893	954	883	1011	1235	1222	0.895	0.912	SPSC
30	1	189	206	220	24	220	222	219	154	198	203	0.859	1.225	SPSC
30	3	307	330	407	125	409	413	404	296	355	355	0.754	1.038	SPSC
30	5	372	421	544	217	541	553	536	411	498	492	0.683	0.904	SPSC
30	10	572	615	775	451	763	790	760	690	855	828	0.738	0.829	SPSC
30	15	741	810	930	663	917	957	913	956	1066	1113	0.796	0.811	SPSC
30	20	890	960	1018	838	1002	1056	1009	1109	1236	1259	0.874	0.888	SPSC
45	1	165	169	188	25	187	189	186	130	172	168	0.881	1.275	SPSC
45	3	271	296	371	118	371	375	369	273	312	294	0.730	0.992	SPSC
45	5	360	389	504	212	506	510	502	376	419	431	0.715	0.957	SPSC
45	10	499	520	675	395	665	684	668	559	646	650	0.739	0.893	SPSC
45	15	651	684	819	577	812	836	811	736	811	808	0.796	0.885	SPSC
45	20	811	836	973	758	954	980	952	901	988	989	0.833	0.900	SPSC
60	1	141	146	165	30	166	166	165	108	134	134	0.858	1.311	SPSC
60	3	229	243	296	117	294	296	292	222	249	239	0.776	1.033	SPSC
60	5	315	319	414	199	412	416	414	320	366	343	0.760	0.985	SPSC
60	10	441	479	605	367	600	608	601	536	564	585	0.729	0.822	SPSC
60	15	588	605	743	524	740	752	738	713	752	736	0.791	0.824	SPSC
60	20	708	753	853	663	845	868	845	838	882	862	0.830	0.845	SPSC
80	1	110	125	118	27	118	118	118	91	111	106	0.931	1.214	SPSC
80	3	231	250	290	116	292	293	290	224	238	239	0.796	1.034	SPSC
80	5	279	295	381	187	379	384	379	293	307	319	0.733	0.954	SPSC
80	10	419	430	542	345	539	545	538	447	493	473	0.772	0.937	SPSC
80	15	534	576	671	486	668	671	667	587	638	655	0.796	0.911	SPSC
80	20	640	663	767	599	761	770	761	691	744	761	0.835	0.927	SPSC
100	1	107	111	112	28	113	113	113	79	86	93	0.954	1.359	SPSC
100	3	192	194	238	107	236	238	236	176	193	201	0.809	1.094	SPSC
100	5	272	273	346	183	343	346	344	266	294	290	0.787	1.024	SPSC
100	10	368	388	471	317	472	472	470	403	438	440	0.780	0.912	SPSC
100	15	466	491	573	430	571	576	572	517	530	535	0.813	0.902	SPSC
100	20	609	634	726	572	725	733	723	661	692	702	0.839	0.921	SPSC

876 G Sensitivity and robustness studies

877 G.1 Probe-rate ablation

878 Sweeping the probe period over $\{5, 10, 20, 30, 50, 100, 300\}$ at $d=4$, $r=1$, $K=4$, $T=6,000$
 879 produces a clearly U-shaped regret curve (Figure 4): sparse probing under-recovers the
 880 subspace; over-frequent probing sacrifices exploitation. Optimum at probe period 20–30,
 881 matching $m_k^* \propto \ell_k^{2/3}$ prediction. The right panel shows a monotone relationship between
 882 late-segment subspace error and final regret, directly linking SPSC’s gains to subspace
 883 recovery quality.

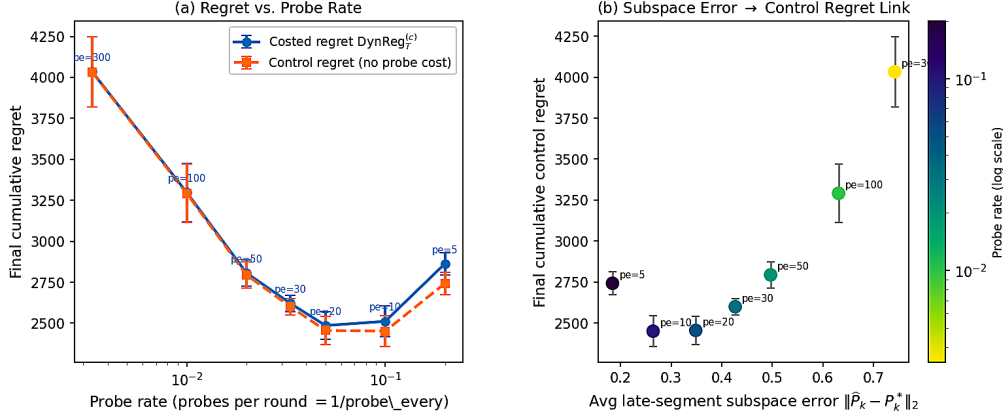


Figure 4: Probe-rate ablation. Left: final regret vs. probe frequency. Right: final control regret vs. late-segment subspace error.

884 G.2 Rank misspecification

885 On Covertypes ($d=155$, true effective rank $r^*=10$, $K=4$, $T=10,000$, 5 seeds), sweeping spec-
 886 ified r at the grid points $\{1, 3, 5, 10, 15, 20, 30, 50, 80\}$ (min $r=1$, max $r=80$) gives the U-
 887 shaped curve in Figure 5. Underestimation loses signal (+33% at $r=1$); mild overestima-
 888 tion captures residual structure (sweet spot at $r=30$, −10%). SPSC beats LinUCB across
 889 $r \in [15, 50]$, so mild overestimation is a safe default.

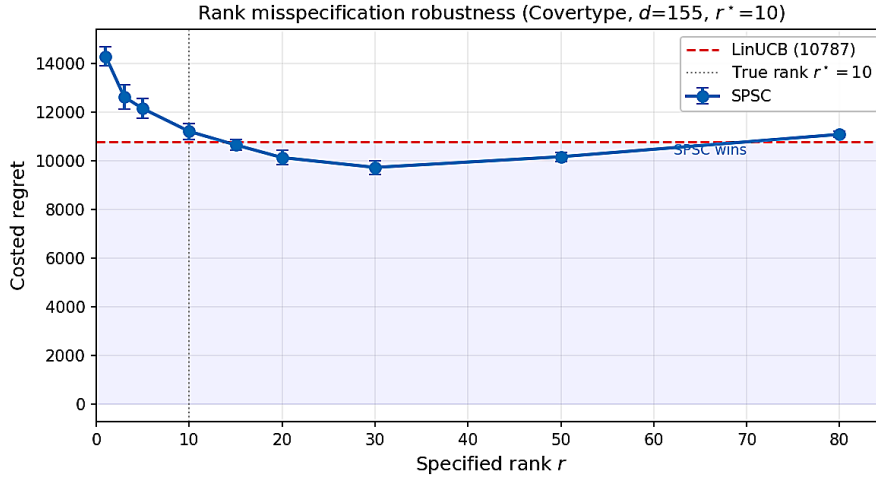


Figure 5: Rank misspecification robustness (Covertypes, $d=155$, $r^*=10$). SPSC beats LinUCB for $r \in [15, 50]$; sweet spot at $r=30$ (10% improvement).

G.3 Robustness and necessity experiments (A/B/C)

Three ablations on the reference setting ($d=4$, $r=1$, $K=4$, $T=6,000$, probe period 30, $W=100$, 10 seeds).

A. Variance misspecification. Injecting $|\delta_\sigma| \in \{0, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$ into the probe statistic $s_t = y_t^2 - (\sigma_\varepsilon^2 + \delta_\sigma)$. Regret and subspace error are stable through $|\delta_\sigma| = 0.5$ (more than $5\times$ the true variance $\sigma_\varepsilon^2 = 0.09$); at maximum misspecification, regret changes by $< 5\%$. For scaled-sphere probes (the theory probe distribution), a constant centering error contributes the population-level identity shift $\tilde{B} = -(\delta_\sigma/d)I_d$, which uniformly shifts eigenvalues without corrupting eigenvectors.

B. Bounded cross-correlation. Injecting $\varepsilon_t \rightarrow \varepsilon_t + \epsilon_\times x_t^\top \theta_t$ for $\epsilon_\times \in \{0, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0\}$. Even at $\epsilon_\times = 1.0$ (cross-correlation equal to signal), regret increases by $< 1\%$, indicating that moderate violations of orthogonality are practically benign.

C. Imperfect probe coverage. Restricting probe directions to the first $d_{\text{cov}} \in \{1, 2, 3, 4\}$ coordinates produces a dramatic monotonic regret blow-up: at $d_{\text{cov}} = 1$, SPSC regret rises to 4,289, nearly matching LinUCB (4,463); subspace error degrades to 0.75. This directly confirms the necessity results: restricted coverage destroys identifiability and forces $\Omega(T)$ regret within the unobserved directions.

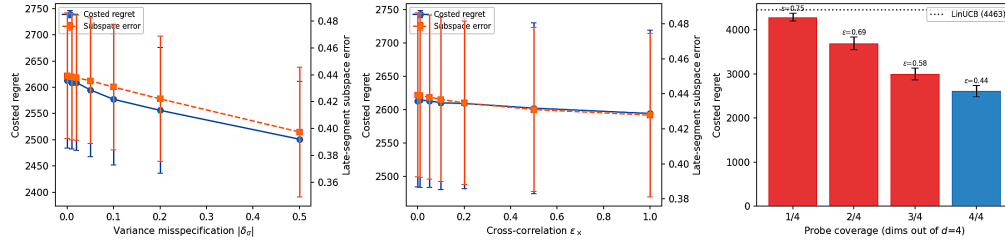


Figure 6: Robustness and necessity experiments. (a) Variance misspecification, regret stable through $|\delta_\sigma| = 0.5$. (b) Cross-correlation relaxation, $< 1\%$ degradation through $\epsilon_\times = 1.0$. (c) Imperfect coverage, regret blow-up with restricted probe directions.

G.4 Assumption-violation master table (large-scale)

We complement the small-scale A/B/C ablations above with a large-scale ($d=60$, $r=5$, $K=10$, $T=5,000$) assumption sweep over (a) variance misspecification $\hat{\sigma}^2/\sigma_\varepsilon^2 \in \{0.5, 1, 2, 4\}$, (b) approximate-rank perturbation $\epsilon_k^\perp \in \{0, 0.05, 0.1, 0.2, 0.4\}$, and (c) spectral radius $\rho(A_k) \in \{0.8, 0.95, 0.99, 1.0\}$ (the last violates the stable-LDS hypothesis). Variance misspec is absorbed into the scaled-identity bias (Lemma B.6) and the relative gap stays flat ($\hat{\sigma}^2/\sigma_\varepsilon^2$ ratio $1.27 \rightarrow 1.16$ as the sweep goes $0.5 \rightarrow 4$); approximate-rank degrades regret near-linearly in $\epsilon_k^\perp$; SPSC remains well-behaved through $\rho = 0.99$ and degrades gracefully at $\rho = 1$ (random-walk limit, outside the stable-LDS assumption).

G.5 Subspace recovery rate and change-point adaptation

Figure 7 plots the binned mean subspace error $\|\hat{P}_k - P_k^*\|_2$ vs. probe count; the empirical rate matches the predicted $1/\sqrt{m}$ on a log-log scale. Figure 8 plots post-change instantaneous regret and the subspace-error trajectory: SPSC recovers substantially faster than ambient LinUCB after each boundary.

G.6 Noise robustness, change-point frequency, drift speed

Three single-parameter sweeps on the reference setting.

Table 17: **Assumption-violation master table** ($d=60, r=5, K=10, T=5,000$, 10 seeds, costed dynamic regret, mean $\pm$ SE). SPSC tracks Oracle through all three sweeps; bold = best non-Oracle in row.

Sweep	Level	Oracle	SPSC-Alg1	SPSC-Adap	LinUCB	SW-LinUCB	D-LinUCB	LowOFUL
$\hat{\sigma}^2/\sigma_\epsilon^2$	0.5	221 $\pm$ 5	281 $\pm$ 10	304 $\pm$ 11	397 $\pm$ 17	395 $\pm$ 16	401 $\pm$ 18	370 $\pm$ 19
	1.0	257 $\pm$ 6	315 $\pm$ 10	319 $\pm$ 10	413 $\pm$ 18	412 $\pm$ 18	415 $\pm$ 19	358 $\pm$ 17
	2.0	299 $\pm$ 10	357 $\pm$ 13	355 $\pm$ 13	421 $\pm$ 19	421 $\pm$ 18	423 $\pm$ 20	377 $\pm$ 14
	4.0	344 $\pm$ 12	398 $\pm$ 16	398 $\pm$ 16	426 $\pm$ 19	425 $\pm$ 19	427 $\pm$ 20	390 $\pm$ 20
$\epsilon_k^\perp$	0.00	257 $\pm$ 6	315 $\pm$ 10	319 $\pm$ 10	413 $\pm$ 18	412 $\pm$ 18	415 $\pm$ 19	358 $\pm$ 17
	0.05	255 $\pm$ 7	311 $\pm$ 12	331 $\pm$ 10	414 $\pm$ 18	414 $\pm$ 18	417 $\pm$ 19	379 $\pm$ 20
	0.10	261 $\pm$ 7	319 $\pm$ 13	331 $\pm$ 12	422 $\pm$ 19	422 $\pm$ 18	425 $\pm$ 19	376 $\pm$ 20
	0.20	286 $\pm$ 8	340 $\pm$ 13	348 $\pm$ 6	452 $\pm$ 19	453 $\pm$ 18	457 $\pm$ 20	399 $\pm$ 19
	0.40	358 $\pm$ 9	396 $\pm$ 10	410 $\pm$ 10	559 $\pm$ 21	561 $\pm$ 20	565 $\pm$ 22	497 $\pm$ 13
$\rho(A_k)$	0.80	713 $\pm$ 6	745 $\pm$ 7	747 $\pm$ 9	744 $\pm$ 8	743 $\pm$ 6	745 $\pm$ 8	732 $\pm$ 9
	0.95	1083 $\pm$ 11	1291 $\pm$ 16	1307 $\pm$ 14	1403 $\pm$ 16	1406 $\pm$ 15	1404 $\pm$ 19	1330 $\pm$ 18
	0.99	1132 $\pm$ 23	2017 $\pm$ 46	1929 $\pm$ 34	2751 $\pm$ 46	2732 $\pm$ 44	2750 $\pm$ 52	2585 $\pm$ 54
	1.00	914 $\pm$ 19	2899 $\pm$ 83	2813 $\pm$ 59	4644 $\pm$ 90	4523 $\pm$ 87	4674 $\pm$ 90	5552 $\pm$ 156

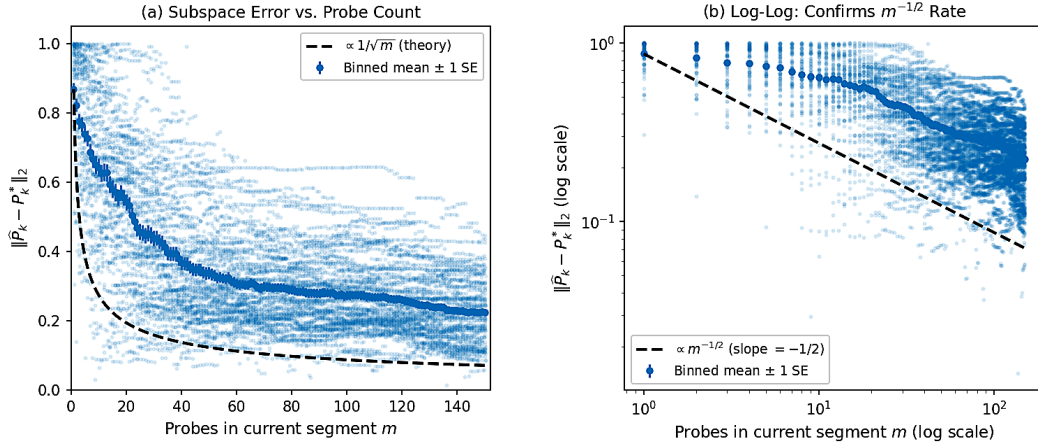


Figure 7: **Subspace recovery.** $\|\hat{P}_k - P_k^*\|_2$ vs. probe count matches the predicted $1/\sqrt{m}$ rate on a log-log scale.

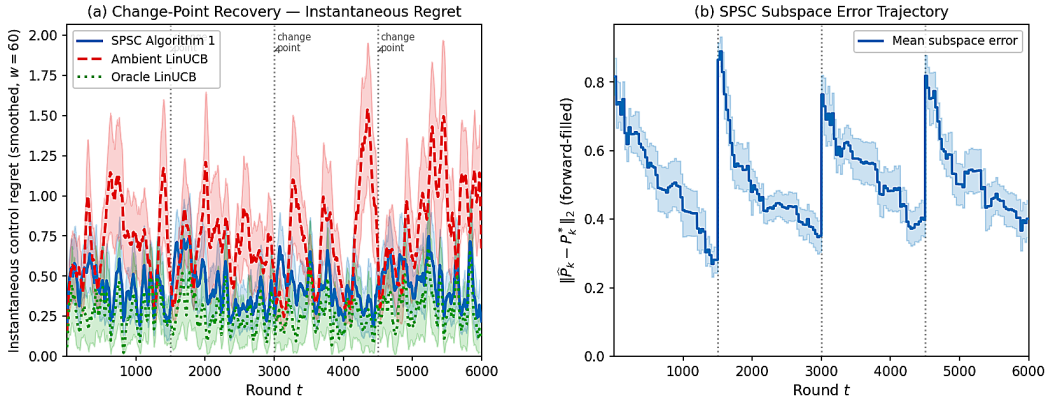


Figure 8: **Change-point adaptation.** Instantaneous regret drops sharply after each boundary as fresh probes rebuild the subspace.

924 **Noise.** Sweeping $\sigma_\varepsilon \in \{0.05, 0.10, 0.20, 0.30, 0.50, 0.80\}$: SPSC/LinUCB ratio grows mildly
 925 with noise (from 0.566 to 0.595); SPSC/Oracle stable at $\approx 1.55\times$. The low-rank benefit is
 926 not a low-noise artifact.

927 **Change-point frequency.** $K \in \{1, 2, 4, 6, 8, 12\}$ at $T=6,000$. Ratio grows from 0.414
 928 (long segments) to 0.855 (very short); SPSC retains an edge at all frequencies but the margin
 929 shrinks as the per-segment probe budget collapses below the identifiability threshold.

930 **Drift speed.** LDS spectral radius $\rho \in \{0.30, \dots, 0.99\}$. Sharp phase transition:
 931 SPSC/LinUCB ≈ 1.01 for $\rho \leq 0.70$, drops to 0.861 at $\rho=0.95$, and to 0.582 at $\rho=0.99$.
 932 Subspace error is nearly constant across ρ (0.28–0.33), so the missing benefit at low ρ is a
 933 signal-energy effect, not estimation failure: low-rank structure in a signal already dominated
 934 by noise leaves little to extract.

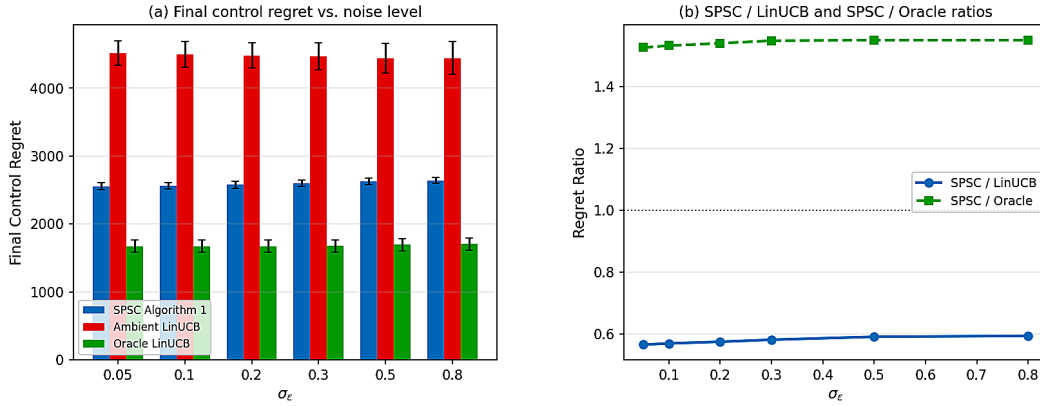


Figure 9: Noise robustness: SPSC advantage is noise-monotone.

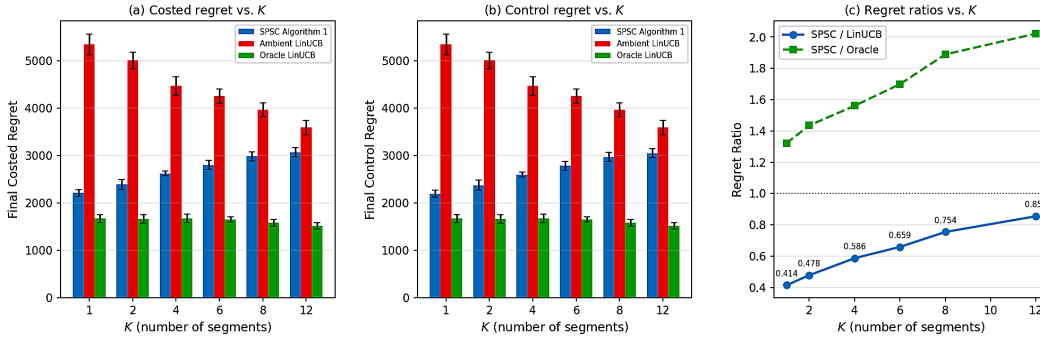


Figure 10: Change-point frequency.

935 H Comparison with nonstationary baselines (Russac benchmark)

936 **Russac et al. NeurIPS 2019 benchmark.** $d=2, r=1, K=4, T=6,000, \sigma_\varepsilon=1$, 50 arms,
 937 30 seeds. SPSC reduces control regret by 46.5% vs. D-LinUCB and 34.8% vs. SW-LinUCB;
 938 probe overhead is a 0.9% fraction.

939 I Warfarin: random-subspace ablation

940 To decompose SPSC’s Warfarin gain into (i) dimensionality reduction and (ii) the *learned*
 941 subspace itself, we replace $\hat{U}_t$ with a fresh random rank- r projection every segment and
 942 run SPSC otherwise unchanged. Results on $d=93$ across $r \in \{1, 2, 3, 5, 10\}$ (10 seeds): the

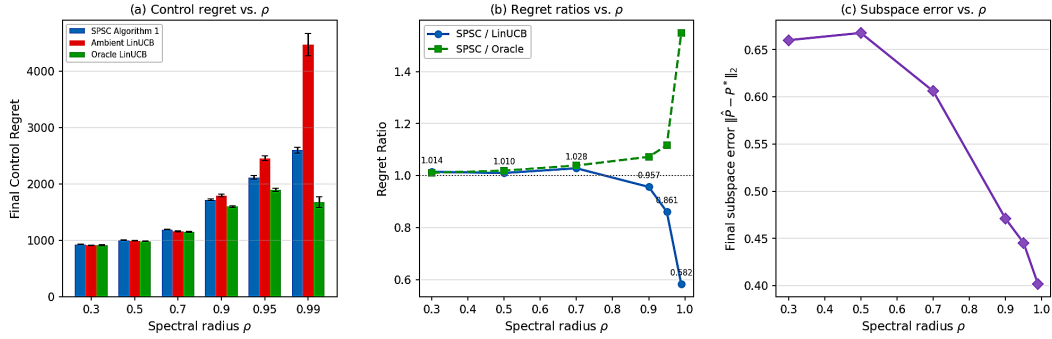


Figure 11: Drift speed. Benefit gated by LDS correlation time $\tau_\rho = 1/(1 - \rho)$.

Table 18: Russac et al. benchmark. Final control regret and costed regret (mean $\pm$ SE, 30 seeds).

Algorithm	Control	Costed	vs. D-LinUCB
SPSC Alg. 1 (ours)	$2,229 \pm 54$	2,249	-46.5%
Oracle LinUCB	$1,757 \pm 55$	1,757	-57.8%
D-LinUCB [Russac et al., 2019]	$4,167 \pm 110$	4,167	—
SW-LinUCB [Cheung et al., 2019]	$3,417 \pm 82$	3,417	-18.0%
OFUL [Abbasi-Yadkori et al., 2011]	$4,645 \pm 149$	4,645	+11.5%
Reset-LinUCB (oracle)	$4,647 \pm 147$	4,647	+11.5%

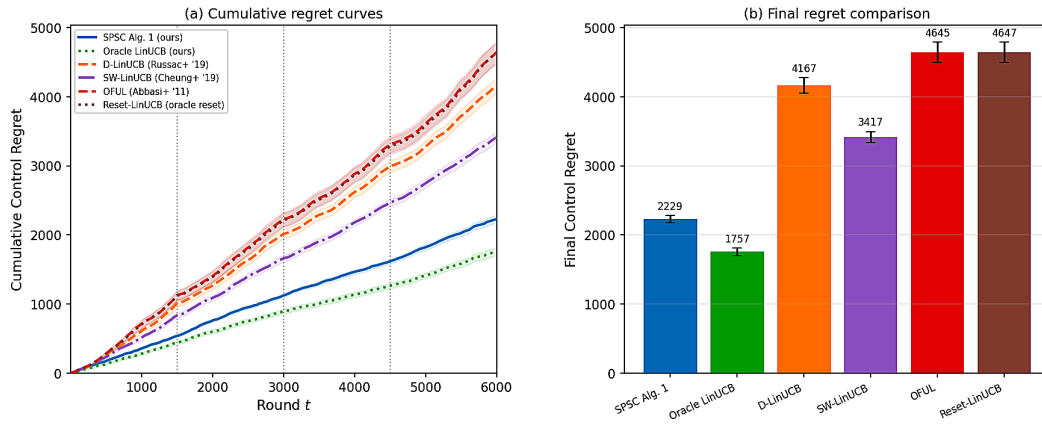


Figure 12: Russac et al. NeurIPS 2019 benchmark.

random-subspace variant retains about half of the SPSC-vs-LinUCB gap, confirming that *dimensionality reduction explains roughly half* the win and the learned subspace explains the rest. The learned subspace captures 2–6 $\times$ more signal variance than a random rank- r projection at every r , directly quantifying the value of the probe-based identification.

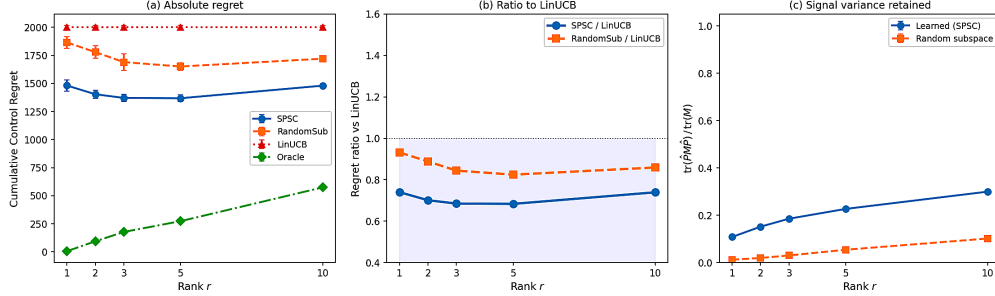


Figure 13: **Warfarin random-subspace ablation.** SPSC’s learned subspace vs. a fresh random rank- r projection at every segment. Half of the gap to LinUCB is dimensionality reduction; the other half is learned signal.

947 J BOSS/Jedra adaptation to the piecewise setting

BOSS [Duong et al., 2024] and Jedra [Jedra et al., 2024] are stationary low-rank methods. To give them a fair comparison to SPSC we restart each at the *true* segment boundaries (an oracle advantage we do not give SPSC) and otherwise leave their hyperparameters at the authors’ recommended defaults. The benchmark is piecewise LDS, $K=10$, $T=5,000$, $\sigma_\varepsilon=0.3$, 40 actions, 10 seeds, probe cost $c=0.1$, on a grid $d \in \{55, 105, 200\} \times r \in \{5, 10, 20\}$ (eight cells: (55, 5), (55, 10), (105, 5), (105, 10), (105, 20), (200, 5), (200, 10), (200, 20)).

Results (cf. Table 4): SPSC Alg. 1 wins every cell, by 5.8–19.9%, with the lead largest at small d (where the ambient d is closest to r and probing pays off most) and narrowing to 5.8–8.0% at $d=200$ as oracle-restart-plus-PCA closes the gap. SPSC-Adaptive (no oracle restarts) also beats BOSS and Jedra on every cell, and the ranking is consistent on every seed. The point is that “low-rank exploitation” alone is not enough for changing-subspace problems: the learner must *identify a changing subspace online*, which is SPSC’s core capability.

961 K Vancomycin: full per-rank tables

This appendix gives the per-rank Vancomycin results ($d=93$, $K=8$, $T=5,000$, $\sigma_\varepsilon=0.3$, 10 seeds, probe period 10, window 200, probe cost $c=0.1$) summarized in Table 3 of the main body. SPSC-Adaptive beats every non-oracle baseline at every rank.

965 L When the adaptive variant wins

This appendix disentangles the two SPSC variants: Alg. 1 assumes oracle change points; Alg. 2 detects them online via the detector statistic (14). When the oracle boundaries are correct, Alg. 1 is slightly better because the adaptive detector pays a small statistical price for unknown change-point localization. When the oracle provides misspecified (false-alarm) boundaries, the picture reverses sharply.

SPSC-Adaptive’s regret is flat as K_{oracle} grows, while Alg. 1 degrades monotonically and is 3 $\times$ worse at $K_{\text{oracle}}=50$: a detector-based reset is strictly safer than a fixed schedule when boundaries are unknown or noisy.

Table 19: **Vancomycin per-rank results** (mean $\pm$ SE, 10 seeds). “vs. best comp.” compares SPSC-Adaptive against the strongest non-oracle baseline in each column (marked *). SPSC-Adaptive wins every rank.

Method	$r=1$	$r=2$	$r=3$	$r=5$	$r=10$
Oracle LinUCB	34.3 ± 1.5	115.1 ± 2.0	197.2 ± 3.0	268.1 ± 3.4	373.2 ± 5.7
SPSC Alg. 1 (ours)	840.3 ± 24.5	805.9 ± 15.4	802.3 ± 12.3	805.3 ± 9.5	811.0 ± 12.2
SPSC-Adaptive	525.3 ± 45.8	613.6 ± 33.1	640.3 ± 33.0	665.9 ± 27.2	705.6 ± 19.5
LowOFUL	1049.9 ± 12.8	$740.1 \pm 36.3^*$	734.5 ± 37.8	791.5 ± 26.0	801.9 ± 20.1
VOFUL	1063.5 ± 5.5	789.7 ± 24.8	$712.5 \pm 25.5^*$	$763.0 \pm 30.6^*$	$758.6 \pm 27.9^*$
LowRank-Reward	1175.6 ± 11.6	806.8 ± 11.2	835.1 ± 10.7	898.8 ± 6.7	880.5 ± 5.5
SW-LinUCB	875.6 ± 6.7	875.6 ± 6.7	875.6 ± 6.7	875.6 ± 6.7	875.6 ± 6.7
D-LinUCB	925.3 ± 5.8	925.3 ± 5.8	925.3 ± 5.8	925.3 ± 5.8	925.3 ± 5.8
Restart-LinUCB	$808.9 \pm 5.3^*$	808.9 ± 5.3	808.9 ± 5.3	808.9 ± 5.3	808.9 ± 5.3
LinUCB	825.3 ± 6.1	825.3 ± 6.1	825.3 ± 6.1	825.3 ± 6.1	825.3 ± 6.1
SPSC-Adaptive vs. LinUCB	-36.4%	-25.7%	-22.4%	-19.3%	-14.5%
SPSC-Adaptive vs. best comp.	-35.1%	-17.1%	-10.1%	-12.7%	-7.0%

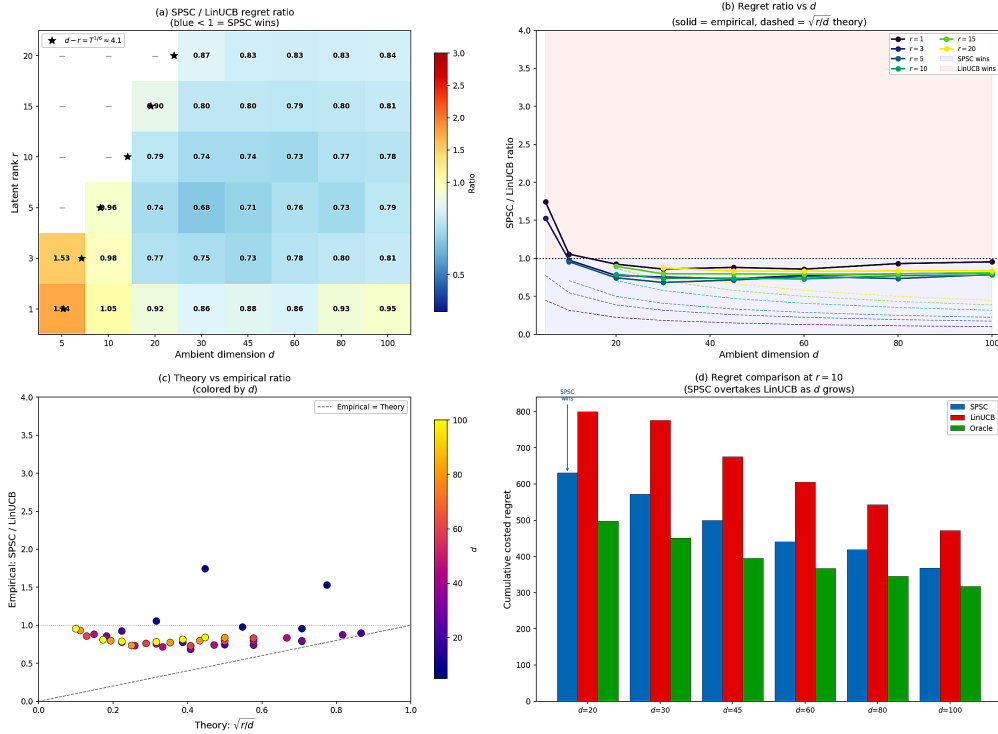


Figure 14: **SPSC-Alg. 1 vs. SPSC-Adaptive**. Top: d -sweep with correct oracle boundaries. Bottom: oracle quality. With $K_{\text{real}}=5$ true boundaries, Alg. 1 is fed $K_{\text{oracle}} \in \{5, \dots, 50\}$ (extras are false alarms) and degrades; Alg. 2 ignores the oracle and is flat.

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Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Limitations are stated where they arise: the regret bound is conditional on the windowed projected potential condition (§4); the bound is most informative when within-segment variation V_{in} is small (Theorem 4.1 discussion); the rule-of-thumb crossover $d - r \gtrsim T^{1/6}$ is qualitative, not sharp (§4, §5.1); SPSC-Adaptive is evaluated empirically only with no formal regret guarantee (§4).

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Justification: All theoretical results (Theorem 2.2, Theorem 4.1, Corollary 4.3, Corollary 4.4, Proposition 4.5) state their assumptions explicitly; full proofs are in Appendix B together with the supporting lemmas (matrix Bernstein, projector concentration, projected nonstationary UCB lemma, prefix concentration).

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Justification: All datasets, dimensions d , ranks r , segment counts K , horizon T , seed counts, probe rate, regularisation, window length, and per-baseline hyperparameters are listed in §5 and Appendix E; per-cell tables in Appendix F report mean $\pm$ SE for every dataset, (d, r) cell, and method.

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Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

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1024 Justification: §5 states the headline parameters per benchmark; Appendix E gives
1025 the full hyperparameter table (probe period, ridge λ , window W , confidence δ ,
1026 probe cost c , baseline-specific knobs) used uniformly across all (d, r) cells without
1027 per-dataset retuning.

1028 7. Experiment statistical significance

1029 Question: Does the paper report error bars suitably and correctly defined or other
1030 appropriate information about the statistical significance of the experiments?

1031 Answer: [Yes]

1032 Justification: Every reported number is $\text{mean} \pm \text{SE}$ or $\text{mean} \pm \text{SEM}$ over at least 10
1033 (typically 10–30) seeds; figures show ± 1 SE bands; the seed count and aggregation
1034 choice are stated in each table caption and in §5.

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1036 Question: For each experiment, does the paper provide sufficient information on the
1037 computer resources (type of compute workers, memory, time of execution) needed
1038 to reproduce the experiments?

1039 Answer: [Yes]

1040 Justification: Appendix E reports that the full grid (all benchmarks, 10 seeds,
1041 10 methods) takes 6–20 hours on a single 16-core CPU; no GPU is required
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1044 Question: Does the research conducted in the paper conform, in every respect, with
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1046 Answer: [Yes]

1047 Justification: The work is a theoretical and empirical study of a stochastic bandit
1048 algorithm on synthetic and publicly available benchmarks; the clinical environ-
1049 ments are semi-synthetic surrogates calibrated to published cohort statistics, with
1050 no patient-identifiable data, no human subjects, and no clinical deployment.

1051 10. Broader impacts

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1054 Answer: [N/A]

1055 Justification: The paper studies a foundational sequential-decision algorithm with
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1057 rank decision problems (recommendation, dosing) and negative-impact pathways
1058 are not direct. Clinical bandit benchmarks are evaluation surrogates, not deployable
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1079 **13. New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The supplemental material releases the SPSC and SPSC-Adaptive Python implementations, the per-benchmark environments, all experiment scripts, and a README documenting the figure-to-script and table-to-source mappings.

1086 **14. Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing or research with human subjects.

1093 **15. Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The paper does not involve human subjects; no IRB approval is required.

1102 **16. Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs are not part of the core methodology; the contributions (algorithm, identification result, regret bound, experiments) are independent of any LLM.